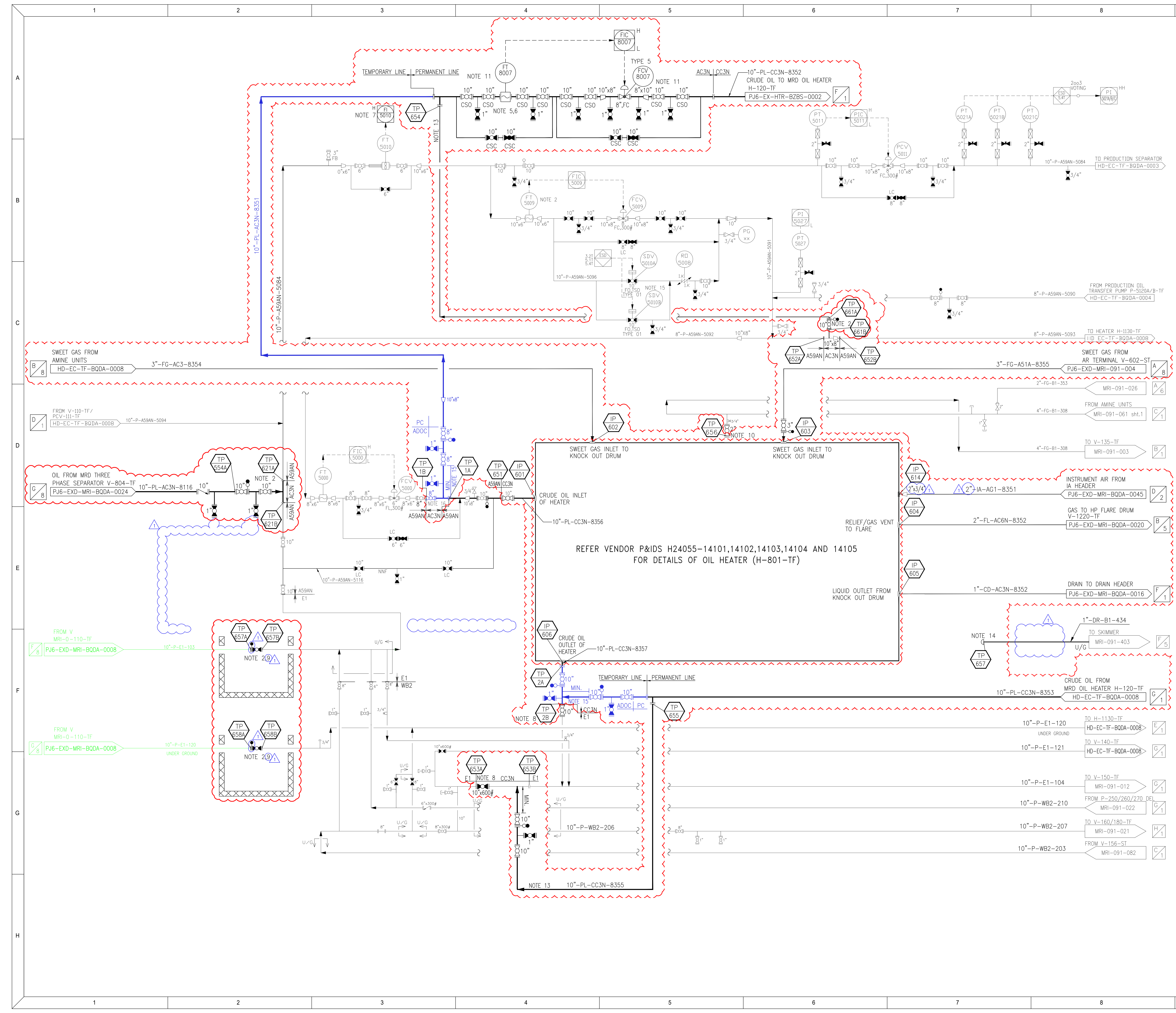




NOTES

- ALL INSTRUMENT TAG NUMBERS ARE SUFFIXED WITH CODE "TF" FOR MUBARRAZ ISLAND.
- EARLY TIE-IN (TP-621A/B, 657A/B, 658A/B, 659A/B, 661A/B) SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.
- HEATER OPERATION WILL BE 2 WORKING AND 1 STANDBY.
- ALL DRAINS SHALL BE CONNECTED TO CLOSED DRAIN SYSTEM AND ALL VENTS SHALL BE CONNECTED TO FLARE.
- ULTRASONIC FLOWMETER.
- MINIMUM STRAIGHT LENGTH REQUIRED AT UPSTREAM AND DOWNSTREAM TO BE 10D AND 5D RESPECTIVELY.
- HIGH ALARM SHALL BE CONFIGURED FOR THE EXISTING FLOW TRANSMITTER FT-5010.
- EXISTING 10" VALVE SHALL BE REUSED.
- DURING COMISIONING THIS VALVE SHALL BE IN OPEN POSITION AND DURING MOTHBALLING PERIOD THIS VALVE SHALL BE CLOSED AND SPADE SHALL BE UNCERTED.
- BLIND FLANGE TO BE ADDED FOR THE EXISTING VALVE.
- FT-8007 AND FCV-8007 TO BE INSTALLED DURING YEAR 2026 SHUTDOWN. HOWEVER, ISOLATION VALVES FOR FCV-8007 AND FT-8007 WILL BE PROVIDED DURING INSTALLATION OF TEMPORARY LINE TO H-120-TF IN YEAR 2025.
- DELETED.
- THIS SPOOL WILL BE CONNECTED DURING YEAR 2026 PLANT SHUTDOWN.
- DRAIN LINE SHALL BE ISOLATED IN THE U/G SECTION WITHOUT AFFECTING THE INSTALLATION OF NEW HEATER H-801-TF.
- MINIMUM TO AVOID DEAR LEG.
- EXISTING 8" VALVE SHALL BE REUSED.

SEQUENCE OF INSTALLATION

YEAR 2025 SHUTDOWN ACTIVITY

- EXISTING HEATER (H-120-TF) WILL BE TAKEN FOR OVERHAULING
- COMPANY WILL INSTALL TIE-IN VALVES FOR CONNECTING TEMPORARY LINES TO NEW HEATER-1 (H-120-TF)

YEAR 2025 POST SHUTDOWN ACTIVITIES WHEN PLANT IS IN OPERATION

- NEW HEATER-1 (H-120-TF) WILL BE INSTALLED AT NEW LOCATION WITHOUT INTERRUPTING THE PLANT OPERATION.
- CONNECT TEMPORARY LINES WITH NEW HEATER-1 (H-120-TF).
- COMMISSION NEW HEATER-1 (H-120-TF) WITH TEMPORARY LINES AND ISOLATE EXISTING HEATER (H-120-TF).
- ONCE NEW HEATER-1 (H-120-TF) OPERATION IS STABILIZED, EXISTING HEATER (H-120-TF) WILL BE DEMOLISHED.

YEAR 2026 SHUTDOWN ACTIVITY

- PERMANENT LINES FOR NEW HEATER-1 (H-120-TF) WILL BE INSTALLED.

YEAR 2026 POST SHUTDOWN ACTIVITIES WHEN PLANT IS IN OPERATION

- RUN NEW HEATER-1 (H-120-TF) WITH PERMANENT LINES WITHOUT INTERRUPTING THE PLANT OPERATION.
- INSTALL NEW HEATER-2 (H-801-TF) AT THE LOCATION WHERE OLD HEATER (H-120-TF) WAS PRESENT. CONNECT OIL, FUEL GAS, FLARE AND DRAIN LINES TO NEW HEATER-2 (H-801-TF).
- NOW NEW HEATER-2 (H-801-TF) IS READY FOR USE.
- DEMOLISH TEMPORARY LINES OF NEW HEATER-1 (H-120-TF).

LEGEND

	SCOPE CLOUD		TIE-IN POINT
	SCOPE CLOUD OF "DETAIL ENGINEERING FOR INSTALLATION OF NEW CRUDE OIL HEATERS (H-120-TF & H-801-TF) AT MUBARRAZ ISLAND" PROJECT.		ISOLATION POINT
	MOTHBALLING SCOPE		TEMPORARY TIE-IN AND LINE
			REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD- MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRF-BQDA-0005
3	P&ID MUBARRAZ ISLAND OIL HEATER (H-120-TF) TRAIN 1 (DEMOLITION)	PJ6-EXD-MRI-BQDA-0029

THIS DRAWING HAS BEEN DEVELOPED FROM DRAWING NO. HD-EC-TF-BQDA-0006 Rev. 2. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

1	30/01/2026	APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	14/11/2025	APPROVED FOR CONSTRUCTION	DVS	ASL	PKR
C	12/09/2025	ISSUED FOR DESIGN	DVS	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	ZAS	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	ZAS	ASL	PKR
REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D

SCALE: NTS	LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002
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ENGINEERING CONSULTANT:
REJLERS
REJLERS TOWER 13TH FLOOR
HAMDAN STREET, PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

COMPANY:
ABU DHABI OIL CO., LTD.

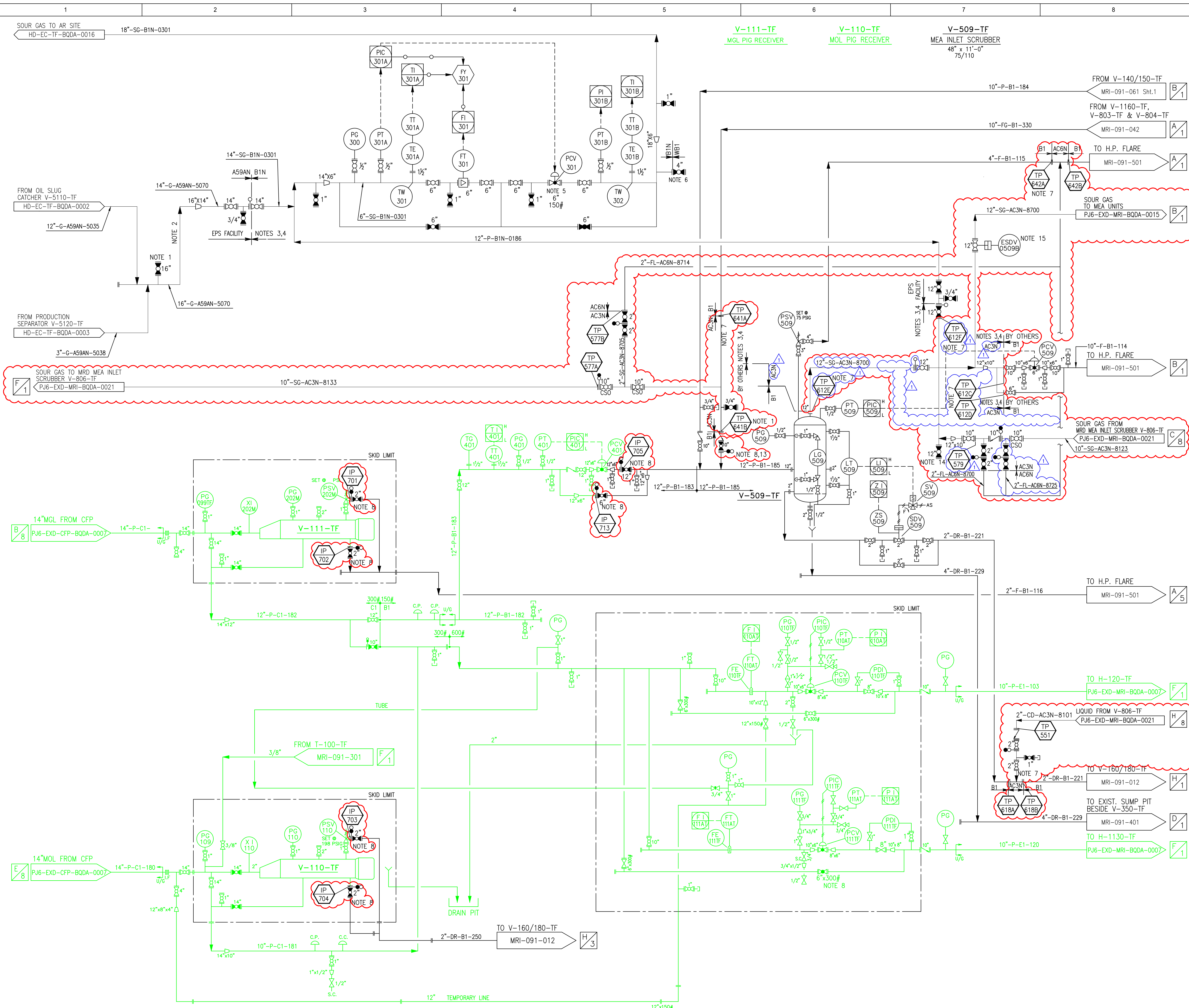
PROJECT:
DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE:
PIPING AND INSTRUMENTATION DIAGRAM
MUBARRAZ ISLAND OIL HEATER (H-801-TF) TRAIN 1

DWG. NO.	PJ6-EXD-MRI-BQDA-0007	REV/NO.	1	SHEET NO.	001
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Filename : PJ6-EXD-MRI-BQDA-0007_1.DWG

SHT. SIZE A1



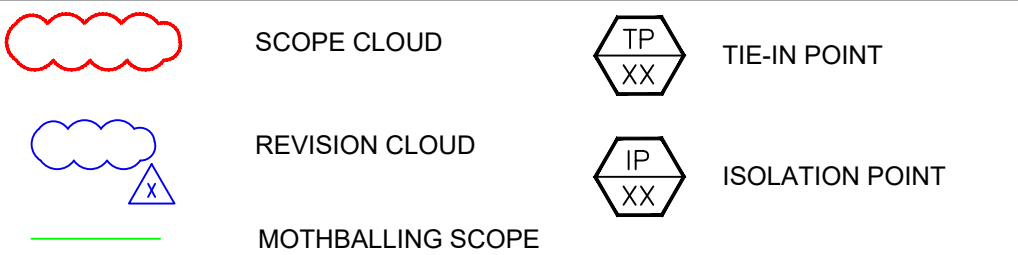
GENERAL NOTES

1. PIPING TIE-IN DETAILS ARRANGEMENTS HAVE BEEN MODIFIED ACCORDING TO LETTER HDS-EPC-L-1-016 PIPING TIE-IN DETAILS AND 8598 -TRAD-ADOC-MM-PM-0065 3RD CONSTRUCTION SITE VISIT WRAP UP MEETING.
2. FOR DEMOLITION SCOPE REFER TO P&ID NO. PJ6-EXD-MRI-BQDA-0030.
3. ONLY ELEMENTS INSIDE CLOUD ARE SCOPE OF HAIL SURFACE FACILITIES EPS PROJECT. REST OF INFO IS SHOWN ONLY FOR INFORMATION (ADOC SCOPE).
4. CALCULATION KIT SHALL BE PROVIDED FOR DISSIMILAR METAL JOINTS.
5. PIPING SPECIFICATION BIN HAS BEEN DEFINED FOR PIPING DESIGN PURPOSE ONLY. THIS SPECIFICATION IS EQUIVALENT TO W2.
6. FOR TIE-IN CONNECTION TYPE, PIPING CLASS AND PIPING DETAIL REFER TO TIE-IN LIST DOC NO. HD-EC-CM-BQOL-0002.
7. FOR AFTER TIE-IN SCOPE REFER TO P&ID HD-EC-TF-BQDA-0042.

NOTES

1. TIE-IN CONNECTION FOR FUTURE FFD COMPRESSOR.
2. NO POCKET.
3. PROCESS DESIGN OF THIS SECTION IS NOT PART OF EPS SCOPE. LINE SIZING/NUMBERING, INSTRUMENT NUMBERING, ISOLATION/DRAINING/VENTING PHILOSOPHY ETC. FOLLOWED AS PER P&ID TO BE PROVIDED BY ADOC (P&ID NO.180-MRI-091-001) INSTRUMENTS ARE PART OF ADOC SCOPE OF SUPPLY.
4. ARRANGEMENT OF THIS SECTION HAS BEEN MODIFIED ACCORDING TO CLIENT ANSWER TO TQ-0045.
5. DELETED.
6. CONTRACTOR TO PROVIDE 4 INCH BRANCH CONNECTION ON 6 INCH SECTION OF LINE (6"-SG-BIN-0301) ALONG WITH 300# FLANGE. BALL VALVE (300#) WITH BLIND FLANGE (300#) SHALL BE PROVIDED BY COMPANY. INSTALLATION OF VALVE FLANGE, ETC, SHALL BE BY CONTRACTOR. DETAILED SCOPE OF CONTRACTOR'S WORK ACCORDING TO LETTER HDS-EPG-CLT-288-4" BRANCH CONNECTION FOR EXISTING SOUR GAS COMPRESSOR NETWORK.
7. EARLY TIE-IN (TP-641A/B, 642A/B, 612C/D/E/F, 618A/B) SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.
8. EXISTING VALVE SHALL BE CLOSED AND SPECTACLE BLIND SHALL BE INSERTED.
ALL VENTS SHALL BE CONNECTED TO FLARE.
10. PLEASE REFER TO MOTHBALLING PHILOSOPHY (Pv6-EXD-GEN-BZOG-0010) FOR DETAILS ON MOTHBALLING SCOPE, INDICATED IN GREEN COLOUR IN THIS P&ID.
DELETED.
12. EXISTING 10" PIPING SHALL BE CHANGED TO 12" PIPING.
13. EXISTING VALVE INTEGRITY TO BE CHECKED; IF NOT OK, THEN THIS HAVE TO BE REPLACED WITH NEW 10" VALVE.
14. FOR FUTURE CONNECTION.
15. M43-ESDV-0509B IS FREE-ISSUED VALVE THROUGH THE ESD PROJECT BY THE ADOC PI-TC TEAM.

LEGEND



REFERENCE DRAWINGS

S. No.	DRAWING/DOC. TITLE	DRAWING/DOC. No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRI-BQDA-0005
3	MOTHBALLING PHILOSOPHY	PJ6-EXD-GEN-BZ0G-0010
4	P&ID MUBARRAZ TERMINAL OIL AND GAS INLET (DEMOLITION)	PJ6-EXD-MRI-BQDA-0030

THIS DRAWING HAS BEEN DEVELOPED FROM FEED
DRAWING NO. HD-EC-TF-BQDA-0009 REV 1. CLOUD
PORTION DENOTES SCOPE OF WORK UNDER DETAILED
ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT
"PJ-2025-CR-002"

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	28/11/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
C	12/09/2025	ISSUED FOR DESIGN	DVS	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	ZAS	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	ZAS	ASL	PKR
REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D
SCALE: NTS		LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. P1-2025-CR-002		

ENGINEERING CONSULTANT:

REJLERS

REJLERS TOWER 13TH FLOOR
HAMDAN STREET, PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

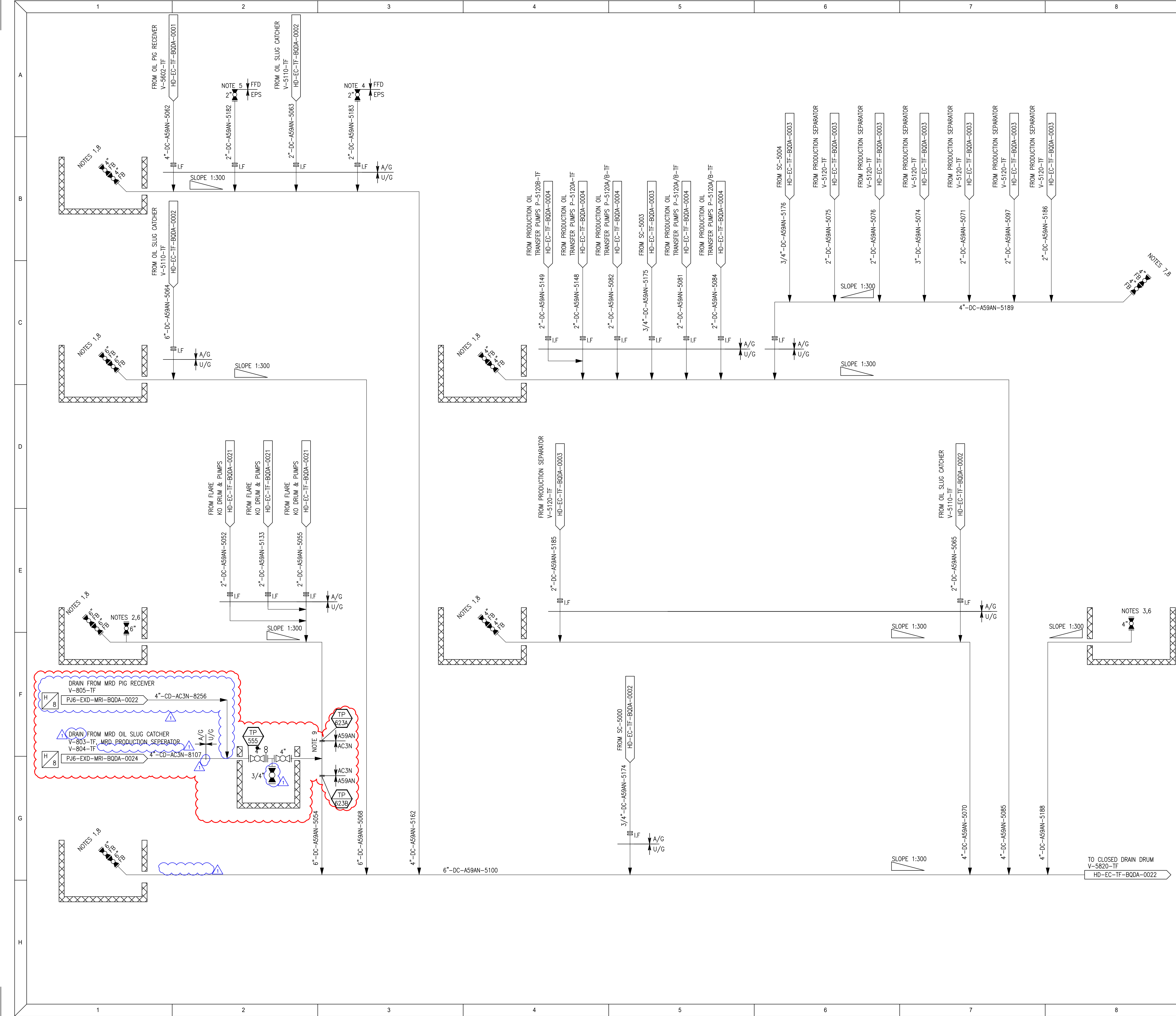
COMPANY:

ABU DHABI OIL CO., LTD.

PROJECT:
DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT
(EPC PHASE)

DRG. TITLE:

DWG. NO.	PJ6-EXD-MRI-BQDA-0008	REV NO.	SHEET NO.
		1	001



- NOTES
1. RODDING FACILITIES TO BE PROVIDED, ALONG WITH CHEQUERED PLATE COVER.

2. FUTURE CONNECTION TO CONNECT NEW CLOSED DRAIN HEADER DURING FFD PHASE.

3. CONNECTION FOR FUTURE PRODUCTION SEPARATOR.

4. BLIND FLANGE TO BE PROVIDED FOR FFD PHASE, DRAIN FROM GAS LIFT PIG LAUNCHER (V-5610-TF).

5. BLIND FLANGE TO BE PROVIDED FOR FFD PHASE, DRAIN FROM GAS INJECTION PIG LAUNCHER (V-5620-F).

6. HAZOP RECOMMENDATION NO.79: FUTURE TIE-INS PROVIDED ON CLOSED DRAIN HEADER SHALL BE LOCATED UNDER GROUND IN VALVE PITS.

7. RODDING FACILITIES TO BE PROVIDED.

8. WATER AND AIR CONNECTION FOR CLOSED DRAIN FLUSHING OPERATIONS TO BE WITHIN 10M OF RODDING OUT POINT.

9. EARLY TIE-IN (TP-623A/B) SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.

10. ALL DRAINS SHALL BE CONNECTED TO CLOSED DRAIN SYSTEM AND ALL VENTS SHALL BE CONNECTED TO FLARE.

LEGEND

SCOPE CLOUD

REVISION CLOUD

TP

XX

TIE-IN POINT

REFERENCE DRAWINGS

S.No.	DRAWING/DOC. TITLE	DRAWING/DOC. No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRI-BQDA-0005
3	P&ID CLOSED DRAIN CONNECTION (DEMOLITION)	PJ6-EXD-MRI-BQDA-0031

THIS DRAWING HAS BEEN DEVELOPED FROM FEED DRAWING NO. HD-EC-TF-BQDA-0017. REV 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	14/11/2025	APPROVED FOR CONSTRUCTION	DVS	ASL	PKR
C	12/09/2025	ISSUED FOR DESIGN	DVS	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	ZAS	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	ZAS	ASL	PKR
REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D

SCALE: NTS	LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002
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ENGINEERING CONSULTANT:

REJLERS

REJLERS TOWER 13TH FLOOR
HAMDAN STREET,PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

COMPANY:

ABU DHABI OIL CO.,LTD.

PROJECT:

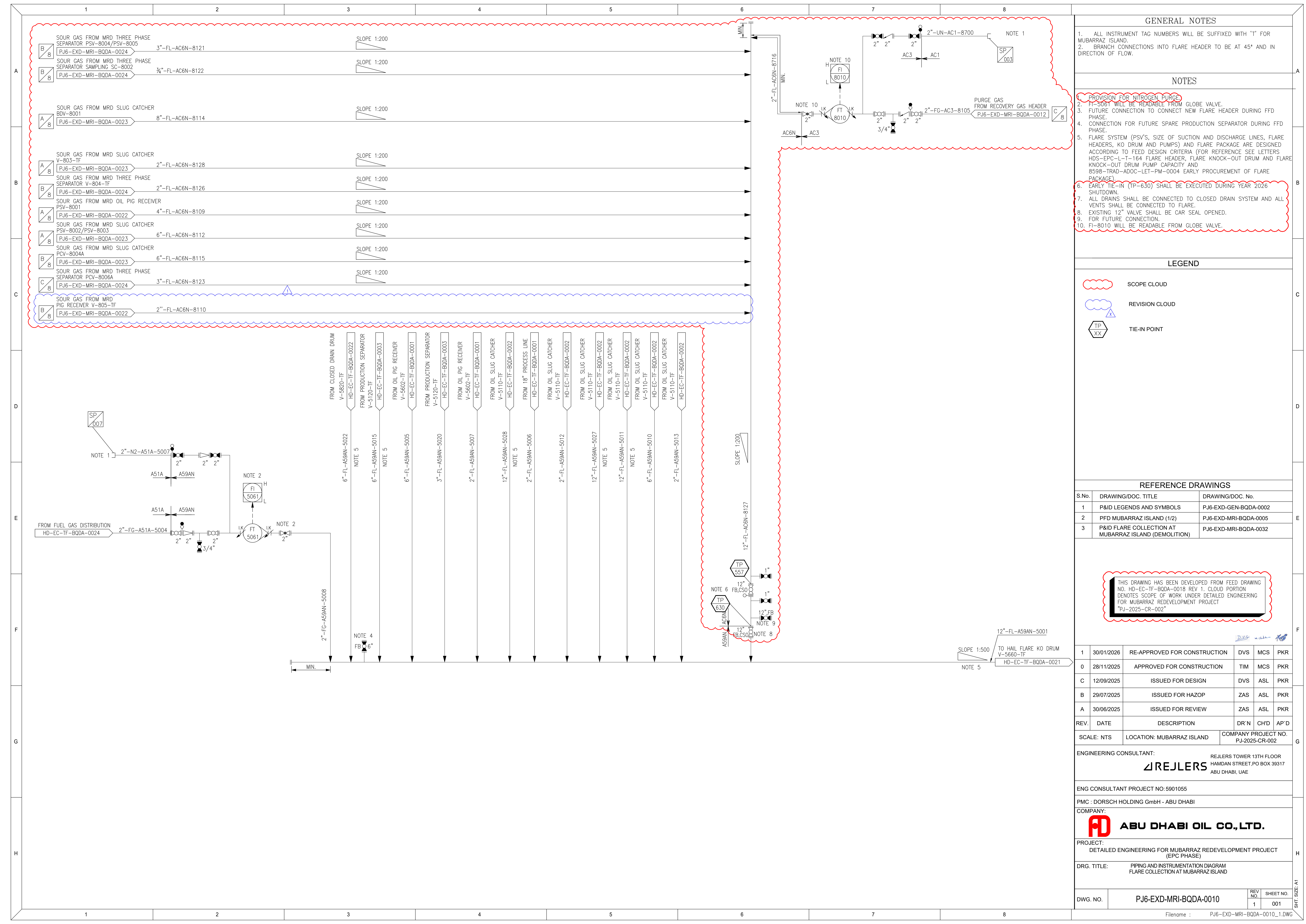
DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE:

PIPING AND INSTRUMENTATION DIAGRAM
CLOSED DRAIN CONNECTION

DWG. NO.	PJ6-EXD-MRI-BQDA-0009	REV NO.	1	SHEET NO.	001
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Filename : PJ6-EXD-MRI-BQDA-0009_1.DWG



GENERAL NOTES

- ALL INSTRUMENT TAG NUMBERS WILL BE SUFFIXED WITH 'I' FOR MUBARRAZ ISLAND.
- BRANCH CONNECTIONS INTO FLARE HEADER TO BE AT 45° AND IN DIRECTION OF FLOW.

NOTES

1. PROVISION FOR NITROGEN PURGE.

2. FI-5061 WILL BE READABLE FROM GLOBE VALVE.

3. FUTURE CONNECTION TO CONNECT NEW FLARE HEADER DURING FFD PHASE.

4. CONNECTION FOR FUTURE SPARE PRODUCTION SEPARATOR DURING FFD PHASE.

5. FLARE SYSTEM (PSV'S, SIZE OF SUCTION AND DISCHARGE LINES, FLARE HEADERS, KO DRUM AND PUMPS) AND FLARE PACKAGE ARE DESIGNED ACCORDING TO FEED DESIGN CRITERIA (FOR REFERENCE SEE LETTERS HDS-EPC-L-T-164 FLARE HEADER, FLARE KNOCK-OUT DRUM AND FLARE KNOCK-OUT DRUM PUMP CAPACITY AND 8598-TRAD-ADOC-LET-PW-0004 EARLY PROCUREMENT OF FLARE PACKAGE).

6. EARLY TIE-IN (TP-630) SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.

7. ALL DRAINS SHALL BE CONNECTED TO CLOSED DRAIN SYSTEM AND ALL VENTS SHALL BE CONNECTED TO FLARE.

8. EXISTING 12" VALVE SHALL BE CAR SEAL OPENED.

9. FOR FUTURE CONNECTION.

10. FI-8010 WILL BE READABLE FROM GLOBE VALVE.

LEGEND

SCOPE CLOUD

REVISION CLOUD

TP
XX

TIE-IN POINT

REFERENCE DRAWINGS

S.No.	DRAWING/DOC. TITLE	DRAWING/DOC. No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRI-BQDA-0005
3	P&ID FLARE COLLECTION AT MUBARRAZ ISLAND (DEMOLITION)	PJ6-EXD-MRI-BQDA-0032

THIS DRAWING HAS BEEN DEVELOPED FROM FEED DRAWING NO. HD-EC-TF-BQDA-0018 REV 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	28/11/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
C	12/09/2025	ISSUED FOR DESIGN	DVS	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	ZAS	ASL	PKR
A	30/06/2025	ISSUED FOR REVIEW	ZAS	ASL	PKR
REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D

SCALE: NTS	LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002
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ENGINEERING CONSULTANT: **REJLERS** REJLERS TOWER 13TH FLOOR
HAMDAN STREET, PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

COMPANY: **ABU DHABI OIL CO.,LTD.**

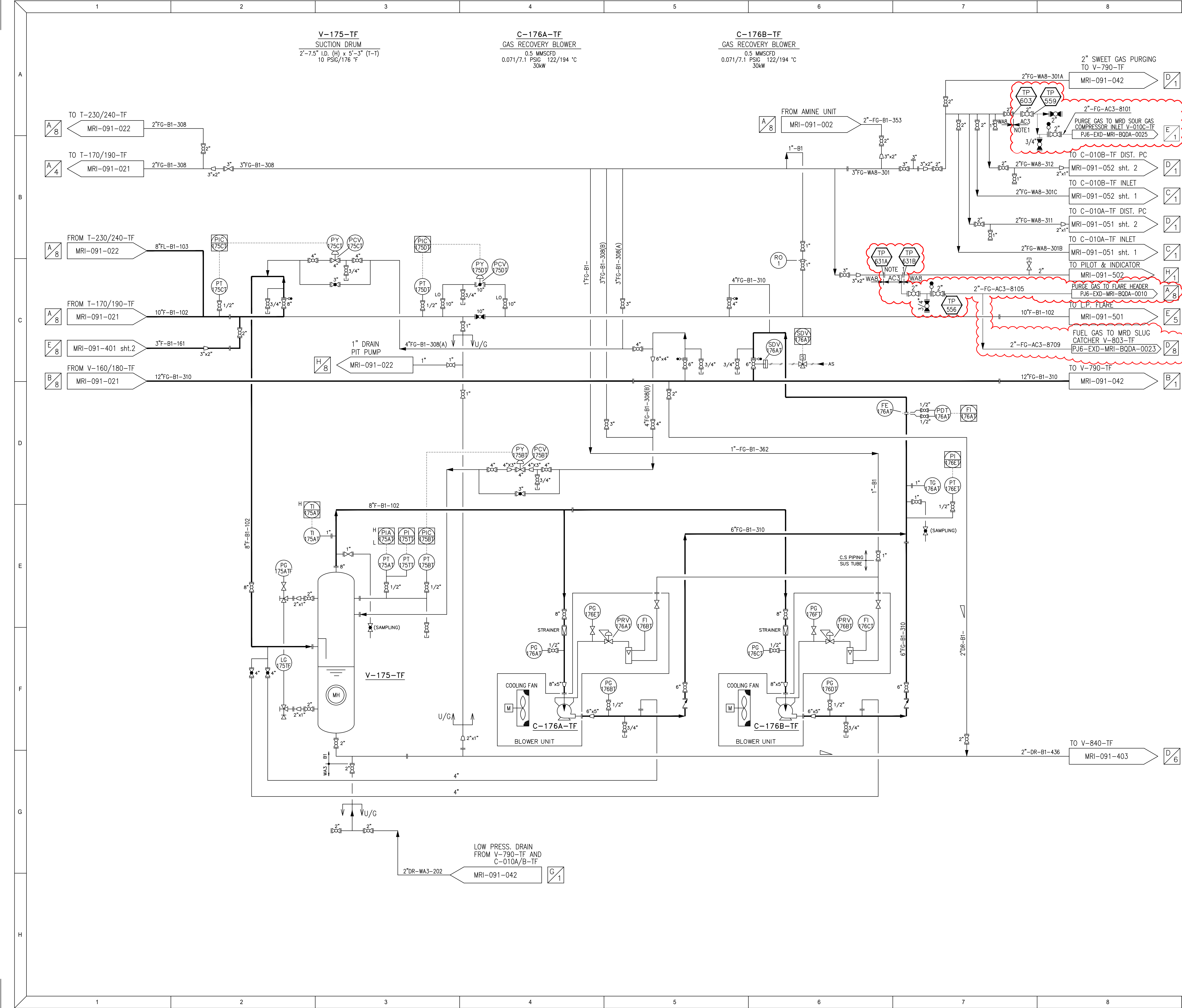
PROJECT: DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE: PIPING AND INSTRUMENTATION DIAGRAM
FLARE COLLECTION AT MUBARRAZ ISLAND

DWG. NO.	PJ6-EXD-MRI-BQDA-0010	REV NO.	1	SHEET NO.	001
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Filename : PJ6-EXD-MRI-BQDA-0010_1.DWG

SHT. SIZE: A1



NOTES

1. EARLY TIE-IN (TP-603, 631A/B) SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.

2. ALL DRAINS SHALL BE CONNECTED TO CLOSED DRAIN SYSTEM AND ALL VENTS SHALL BE CONNECTED TO FLARE.

LEGEND

SCOPE CLOUD

REVISION CLOUD

TP
XX

TIE-IN POINT

REFERENCE DRAWINGS

S.No.	DRAWING/DOC. TITLE	DRAWING/DOC. No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRI-BQDA-0005
3	P&ID MUBARRAZ TERMINAL TANK GAS RECOVERY SYSTEM (DEMOLITION)	PJ6-EXD-MRI-BQDA-0034

THIS DRAWING HAS BEEN DEVELOPED FROM FEED DRAWING NO. MRI-091-026_REV 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D
1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	14/11/2025	APPROVED FOR CONSTRUCTION	DVS	ASL	PKR
C	12/09/2025	ISSUED FOR DESIGN	DVS	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	ZAS	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	ZAS	ASL	PKR

SCALE:	LOCATION:	COMPANY PROJECT NO.
NTS	MUBARRAZ ISLAND	PJ-2025-CR-002

ENGINEERING CONSULTANT:

AREJLERS

REJILERS TOWER 13TH FLOOR
HAMDAN STREET, PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

COMPANY:

ABU DHABI OIL CO.,LTD.

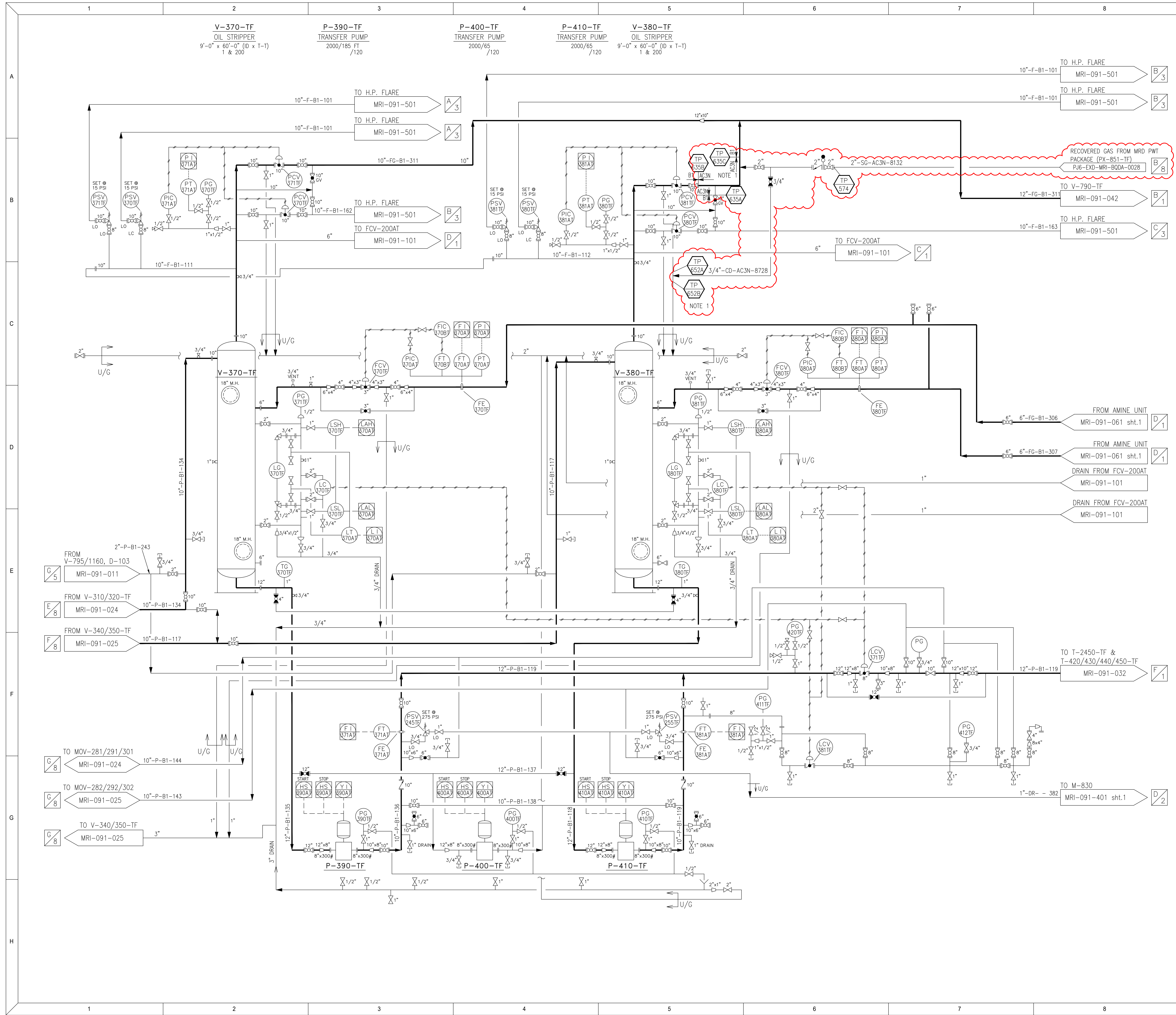
PROJECT:
DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE:
PIPING AND INSTRUMENTATION DIAGRAM
MUBARRAZ TERMINAL
TANK GAS RECOVERY SYSTEM

DWG. NO.	PJ6-EXD-MRI-BQDA-0012	REV NO.	1	SHEET NO.	001
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Filename : PJ6-EXD-MRI-BQDA-0012_1.DWG

SHEET SIZE A1



NOTES

1. EARLY TIE-IN (TP-635A/B/C,652A/B) SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.

LEGEND

SCOPE CLOUD

REVISION CLOUD

TIE-IN POINT

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGEND AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (2/2)	PJ6-EXD-MRI-BQDA-0005
3	P&ID MUBARRAZ TERMINAL - OIL STRIPPER (DEMOLITION)	PJ6-EXD-MRI-BQDA-0035

THIS DRAWING HAS BEEN DEVELOPED FROM DRAWING NO.
MRI-091-031 REV 0. CLOUDED PORTION DENOTES
SCOPE OF WORK UNDER DETAILED ENGINEERING FOR
MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PH
0	14/11/2025	APPROVED FOR CONSTRUCTION	DVS	ASL	PH
C	12/09/2025	ISSUED FOR DESIGN	DVS	ASL	PH
B	29/07/2025	ISSUED FOR HAZOP	ZAS	ASL	PH
A	03/07/2025	ISSUED FOR REVIEW	ZAS	ASL	PH
REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D

SCALE: NTS	LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO PJ-2025-CR-002
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ENGINEERING CONSULTANT:

REJLERS

REJLERS TOWER 13TH FLOOR
HAMDAN STREET, PO BOX 3931
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

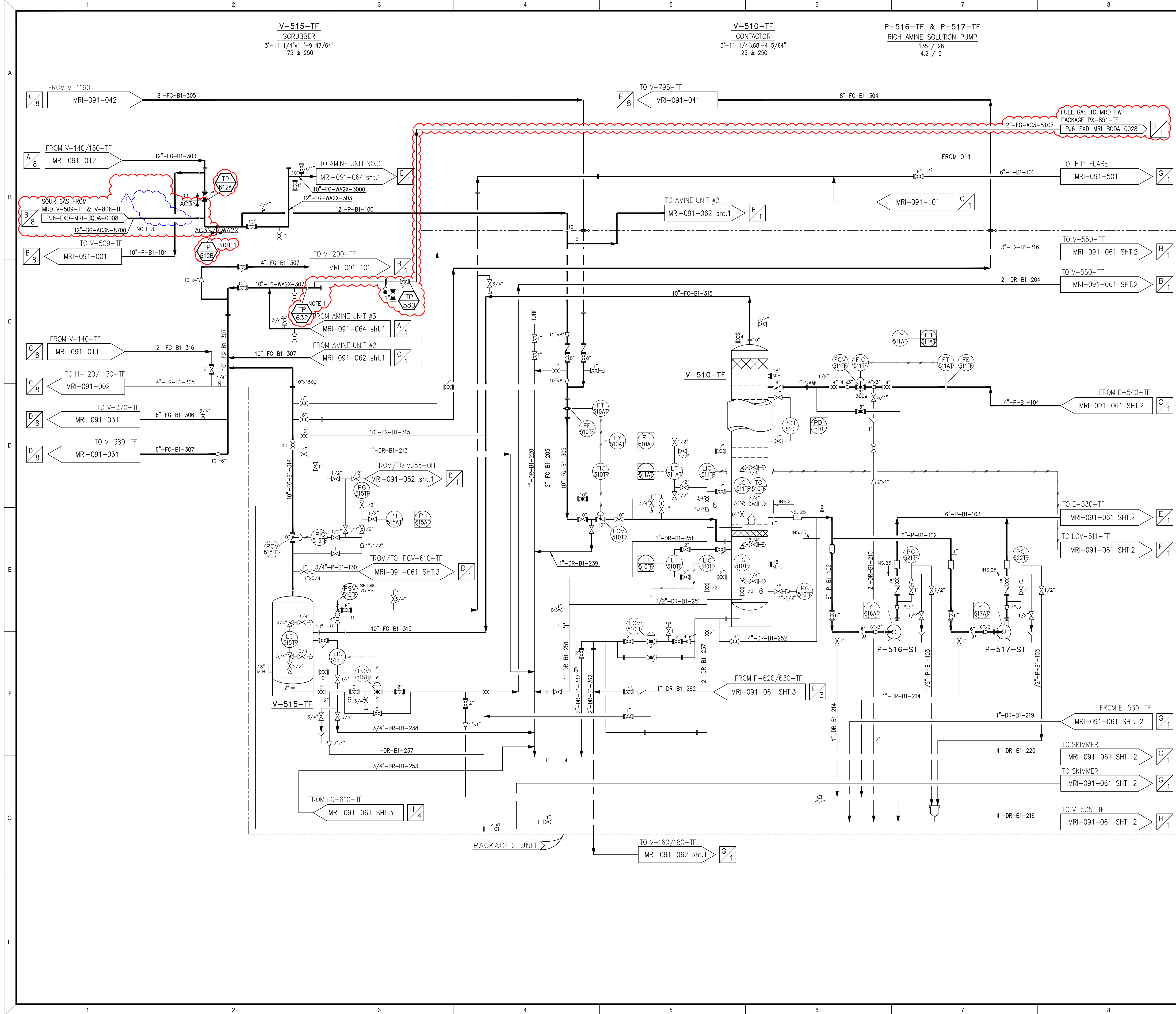
COMPANY: _____

PROJECT:
DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT
(EPC PHASE)

ORG. TITLE: PIPING AND INSTRUMENTATION DIAGRAM
MUBARRAZ TERMINAL
OIL STRIPPER

DWG. NO.	PJ6-EXD-MRI-BQDA-0013	REV NO.	SHEET NO.
		1	001

Filename : PJ6-EXD-MRI-BQDA-0013_1.DWG



NOTES

1. EARLY TIE-IN (TP-612A/B, TP-632) SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.
2. DELETED.
3. EXISTING 10" PIPING SHALL BE CHANGED TO 12" PIPING.

LEGEND

SCOPE CLOUD

REVISION CLOUD

TIE-IN POINT

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-MRF-BQDA-0050
2	PFID MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRI-BQDA-0005
3	P&ID MUBARRAZ TERMINAL AMINE UNIT #1 (1 OF 4) (DEMOLITION)	PJ6-EXD-MRI-BQDA-0043

THIS DRAWING HAS BEEN DEVELOPED FROM FEED DRAWING NO. MRI-091-061 REV 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	28/11/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
C	12/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	ZAS	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	ZAS	ASL	PKR
REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D

SCALE: NTS LOCATION: MUBARRAZ ISLAND COMPANY PROJECT NO. PJ-2025-CR-002

ENGINEERING CONSULTANT: **REJLERS** REJLERS TOWER 13TH FLOOR, HAMDAN STREET, PO BOX 39317, ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC: DORSCH HOLDING GmbH - ABU DHABI

COMPANY: **ABU DHABI OIL CO., LTD.**

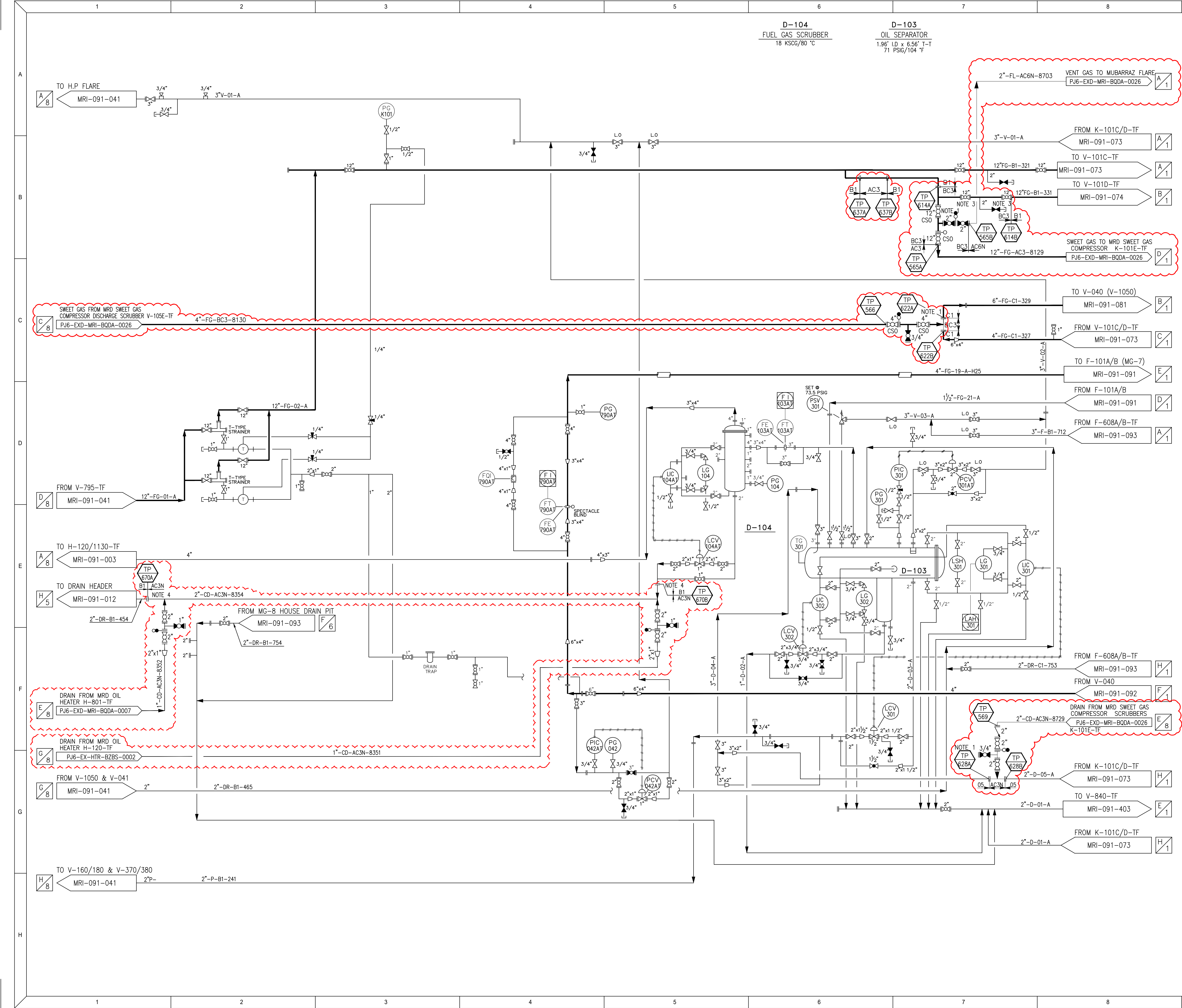
PROJECT: DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE: PIPING AND INSTRUMENTATION DIAGRAM MUBARRAZ TERMINAL AMINE UNIT #1 (1 OF 4)

DWG. NO. PJ6-EXD-MRI-BQDA-0015

REV. NO. 1 SHEET NO. 001

Filename : PJ6-EXD-MRI-BQDA-0015_1.DWG



NOTES

- EARLY TIE-IN (TP-622A/B, TP-628A/B) SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.
- ALL DRAINS SHALL BE CONNECTED TO CLOSED DRAIN SYSTEM AND ALL VENTS SHALL BE CONNECTED TO FLARE.
- EXISTING 12" VALVES SHALL BE RELOCATED FROM TP-637A/B.
- FLANGES AS PER 3D MODEL

SEQUENCE OF INSTALLATION

YEAR 2025 SHUTDOWN ACTIVITY

- EXISTING HEATER (H-120-TF) WILL BE TAKEN FOR OVERHAULING
- COMPANY WILL INSTALL TIE-IN VALVES FOR CONNECTING TEMPORARY LINES TO NEW HEATER-1 (H-120-TF)

YEAR 2025 POST SHUTDOWN ACTIVITIES WHEN PLANT IS IN OPERATION

- NEW HEATER-1 (H-120-TF) WILL BE INSTALLED AT NEW LOCATION WITHOUT INTERRUPTING THE PLANT OPERATION.
- CONNECT TEMPORARY LINES WITH NEW HEATER-1 (H-120-TF).
- COMMISSION NEW HEATER-1 (H-120-TF) WITH TEMPORARY LINES AND ISOLATE EXISTING HEATER (H-120-TF).
- ONCE NEW HEATER-1 (H-120-TF) OPERATION IS STABILIZED, EXISTING HEATER (H-120-TF) WILL BE DEMOLISHED.

YEAR 2026 SHUTDOWN ACTIVITY

- PERMANENT LINES FOR NEW HEATER-1 (H-120-TF) WILL BE INSTALLED.
- RUN NEW HEATER-1 (H-120-TF) WITH PERMANENT LINES WITHOUT INTERRUPTING THE PLANT OPERATION.
- INSTALL NEW HEATER-2 (H-801-TF) AT THE LOCATION WHERE OLD HEATER (H-120-TF) WAS PRESENT, CONNECT OIL, FUEL GAS, FLARE AND DRAIN LINES TO NEW HEATER-2 (H-801-TF).
- NOW NEW HEATER-2 (H-801-TF) IS READY FOR USE.
- DEMOLISH TEMPORARY LINES OF NEW HEATER-1 (H-120-TF)

LEGEND

SCOPE CLOUD

SCOPE CLOUD OF "DETAIL ENGINEERING FOR INSTALLATION OF NEW CRUDE OIL HEATERS (H-120-TF & H-801-TF) AT MUBARRAZ ISLAND" PROJECT.

TP XX

TIE-IN POINT

REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS		
S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PPD MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRF-BQDA-0005
3	P&ID MUBARRAZ TERMINAL SWEET GAS FLOW & VESSEL D103 (DEMOLITION)	PJ6-EXD-MRI-BQDA-0037

THIS DRAWING HAS BEEN DEVELOPED FROM DRAWING NO. MRI-091-071 Rev 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

REV.	DATE	DESCRIPTION	DR'N	CHD	AP'D
1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
0	28/11/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
C	12/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	ZAS	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	ZAS	ASL	PKR

SCALE:	LOCATION:	COMPANY PROJECT NO.
NTS	MUBARRAZ ISLAND	PJ-2025-CR-002

ENGINEERING CONSULTANT:

AREJLERS

REJILERS TOWER 13TH FLOOR
HAMDAN STREET, PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

COMPANY:

ABU DHABI OIL CO.,LTD.

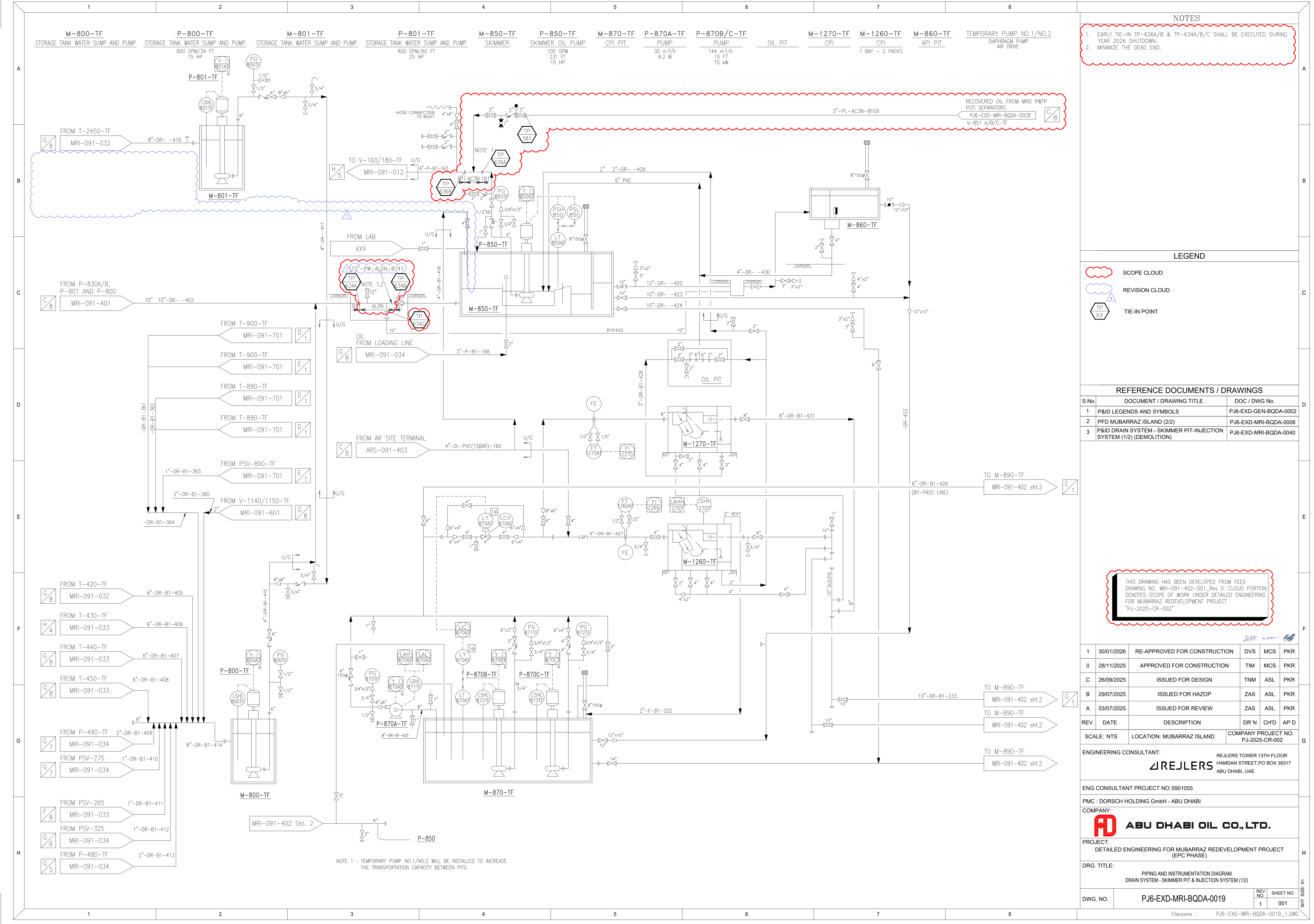
PROJECT:
DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE:
PIPING AND INSTRUMENTATION DIAGRAM
MUBARRAZ TERMINAL SWEET GAS FLOW & VESSEL D103

DWG. NO.	PJ6-EXD-MRI-BQDA-0016	REV NO.	1	SHEET NO.	001
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Filename : PJ6-EXD-MRI-BQDA-0016_1.DWG

SHR SIZE A1



- NOTES
1. EARLY TIE-IN TP-636A/B & TP-634A/B/C SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.
 2. MINIMIZE THE DEAD END.

- LEGEND
- SCOPE CLOUD
 - REVISION CLOUD
 - TIE-IN POINT

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (2/2)	PJ6-EXD-MRI-BQDA-0006
3	P&ID DRAIN SYSTEM - SKIMMER PIT-INJECTION SYSTEM (1/2) (DEMOLITION)	PJ6-EXD-MRI-BQDA-0040

THIS DRAWING HAS BEEN DEVELOPED FROM FEED DRAWING NO. MRI-091-402-001_Rev 0. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	28/11/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
C	26/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	ZAS	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	ZAS	ASL	PKR
REV.	DATE	DESCRIPTION	DR 'N	CH'D	AP'D

ENGINEERING CONSULTANT: **IREJLERS** REJILERS TOWER 13TH FLOOR HAMDAN STREET, PO BOX 39317 ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

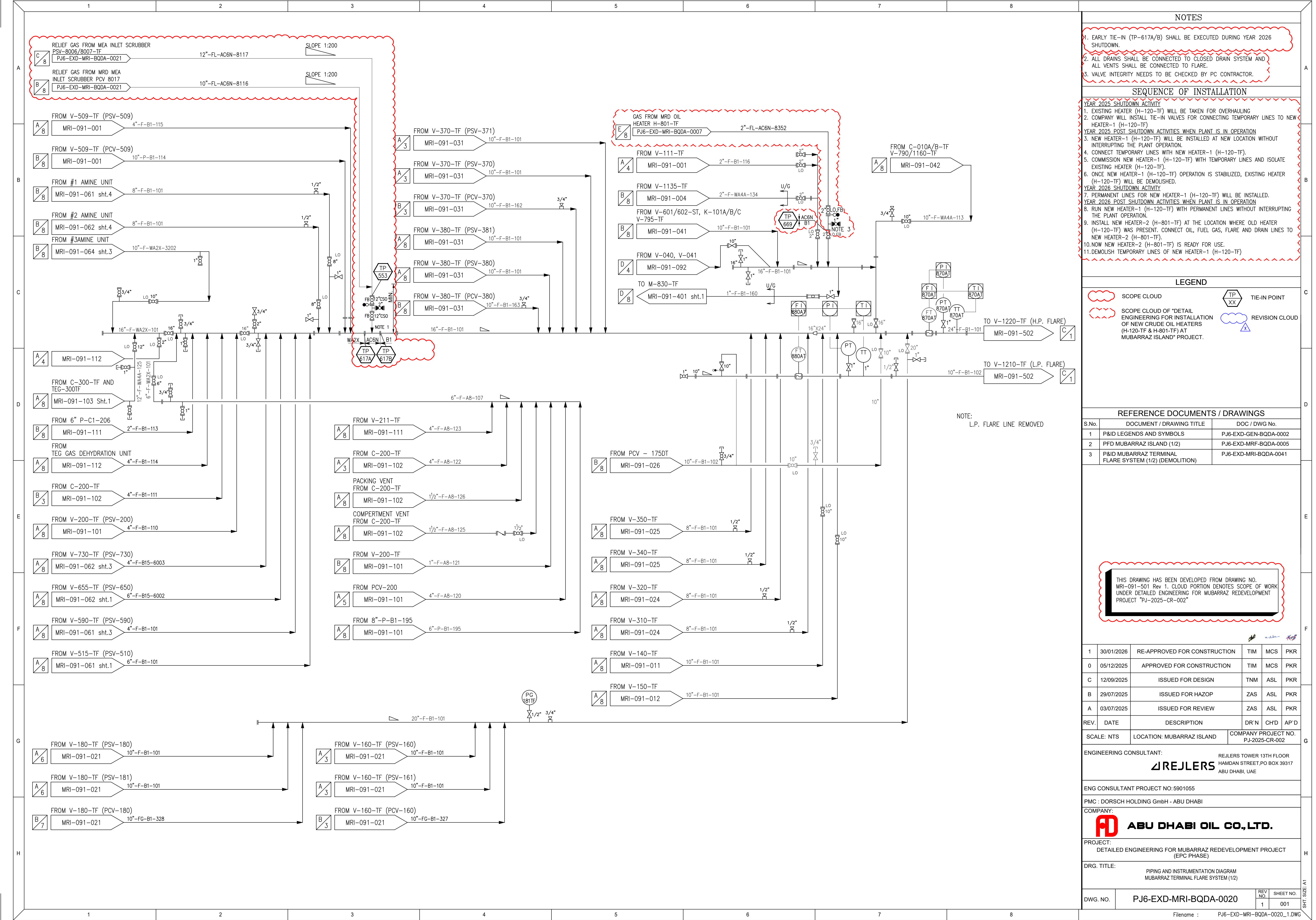
PMC : DORSCH HOLDING GmbH - ABU DHABI

COMPANY: **ABU DHABI OIL CO., LTD.**

PROJECT: DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE: PIPING AND INSTRUMENTATION DIAGRAM DRAIN SYSTEM - SKIMMER PIT & INJECTION SYSTEM (1/2)

DWG. NO.	PJ6-EXD-MRI-BQDA-0019	REV NO.	1	SHEET NO.	001
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- NOTES
1. EARLY TIE-IN (TP-617A/B) SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.

2. ALL DRAINS SHALL BE CONNECTED TO CLOSED DRAIN SYSTEM AND ALL VENTS SHALL BE CONNECTED TO FLARE.

3. VALVE INTEGRITY NEEDS TO BE CHECKED BY PC CONTRACTOR.

- SEQUENCE OF INSTALLATION
- YEAR 2025 SHUTDOWN ACTIVITY

1. EXISTING HEATER (H-120-TF) WILL BE TAKEN FOR OVERHAULING

2. COMPANY WILL INSTALL TIE-IN VALVES FOR CONNECTING TEMPORARY LINES TO NEW HEATER-1 (H-120-TF)

YEAR 2025 POST SHUTDOWN ACTIVITIES WHEN PLANT IS IN OPERATION

3. NEW HEATER-1 (H-120-TF) WILL BE INSTALLED AT NEW LOCATION WITHOUT INTERRUPTING THE PLANT OPERATION.

4. CONNECT TEMPORARY LINES WITH NEW HEATER-1 (H-120-TF).

5. COMMISSION NEW HEATER-1 (H-120-TF) WITH TEMPORARY LINES AND ISOLATE EXISTING HEATER (H-120-TF).

6. ONCE NEW HEATER-1 (H-120-TF) OPERATION IS STABILIZED, EXISTING HEATER (H-120-TF) WILL BE DEMOLISHED.

YEAR 2026 SHUTDOWN ACTIVITY

7. PERMANENT LINES FOR NEW HEATER-1 (H-120-TF) WILL BE INSTALLED.

YEAR 2026 POST SHUTDOWN ACTIVITIES WHEN PLANT IS IN OPERATION

8. RUN NEW HEATER-1 (H-120-TF) WITH PERMANENT LINES WITHOUT INTERRUPTING THE PLANT OPERATION.

9. INSTALL NEW HEATER-2 (H-801-TF) AT THE LOCATION WHERE OLD HEATER (H-120-TF) WAS PRESENT. CONNECT OIL, FUEL GAS, FLARE AND DRAIN LINES TO NEW HEATER-2 (H-801-TF).

10. NOW NEW HEATER-2 (H-801-TF) IS READY FOR USE.

11. DEMOLISH TEMPORARY LINES OF NEW HEATER-1 (H-120-TF)

- LEGEND
- SCOPE CLOUD

SCOPE CLOUD OF "DETAIL ENGINEERING FOR INSTALLATION OF NEW CRUDE OIL HEATERS (H-120-TF & H-801-TF) AT "MUBARRAZ ISLAND" PROJECT.
- TP

XX

TIE-IN POINT
- REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFID MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRF-BQDA-0005
3	P&ID MUBARRAZ TERMINAL FLARE SYSTEM (1/2) (DEMOLITION)	PJ6-EXD-MRI-BQDA-0041

THIS DRAWING HAS BEEN DEVELOPED FROM DRAWING NO. MRI-091-501 Rev 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
0	05/12/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
C	12/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	ZAS	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	ZAS	ASL	PKR
REV.	DATE	DESCRIPTION	DR'N	CHD	AP'D

SCALE: NTS	LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002
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ENGINEERING CONSULTANT:

AREJLERS

REJLERS TOWER 13TH FLOOR
HAMDAN STREET,PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

COMPANY:

 **ABU DHABI OIL CO.,LTD.**

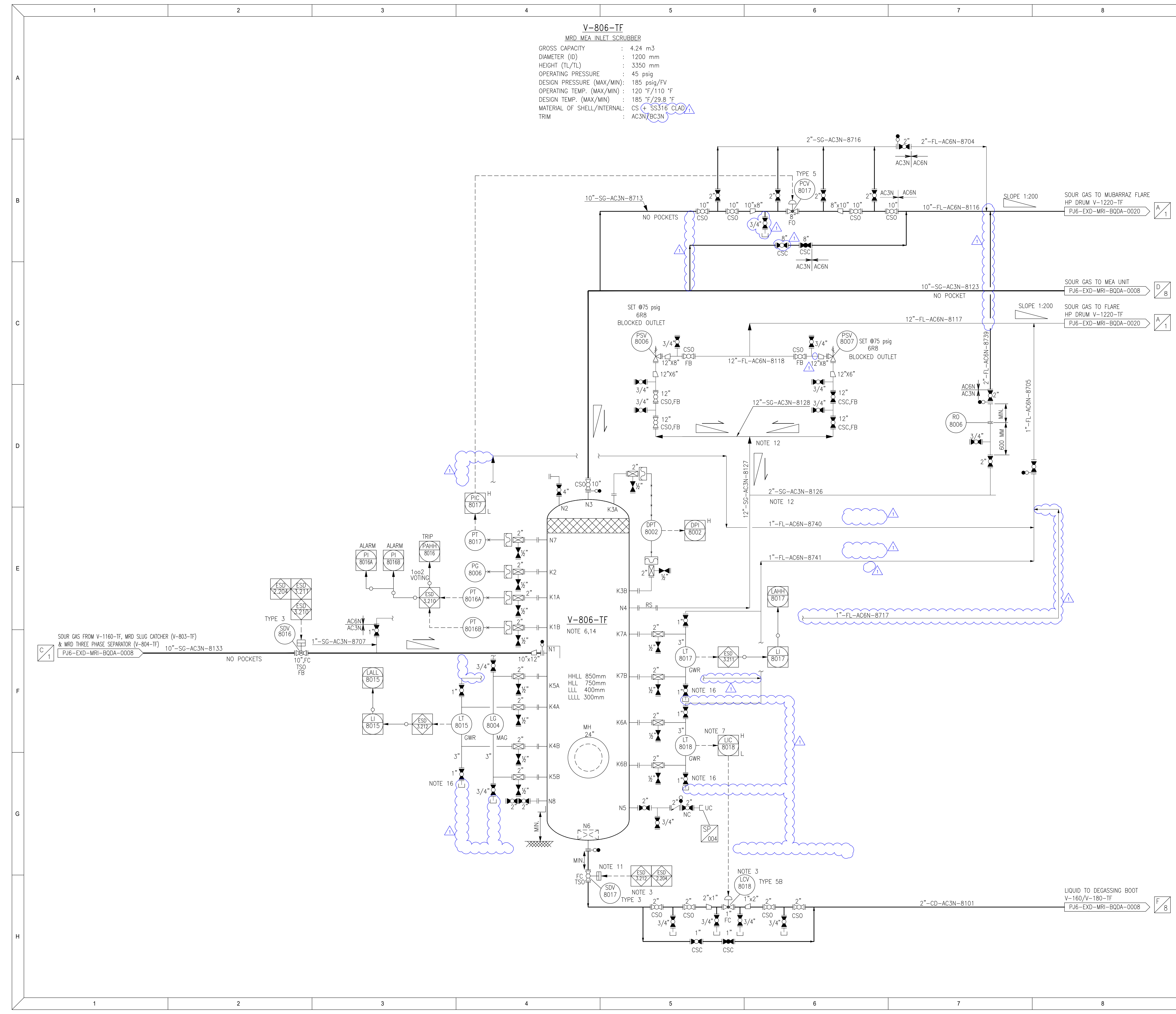
PROJECT:

DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE:

PIPING AND INSTRUMENTATION DIAGRAM
MUBARRAZ TERMINAL FLARE SYSTEM (1/2)

DWG. NO.	PJ6-EXD-MRI-BQDA-0020	REV NO.	1	SHEET NO.	001
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NOTES

- ALL INSTRUMENT TAG NUMBERS SHALL BE SUFFIXED BY SITE SYMBOL CODE "TF" FOR MUBARRAZ ISLAND.
- DELETED.
- SDV-8017 AND LCV-8018 TO BE LOCATED AT MINIMUM DISTANCE FROM V-806-TF.
- ALL DRAINS SHALL BE CONNECTED TO CLOSED DRAIN SYSTEM AND ALL VENTS SHALL BE CONNECTED TO FLARE.
- FUEL GAS SYSTEM IS DESIGNED FOR COMPLETE GAS FROM MUBARRAZ ISLAND.
- DEMISTER PAD SHALL BE DESIGN TO REMOVE 99.5% OF DROPLETS ABOVE 10 MICRONS.
- MODULATING LEVEL CONTROL IS CONSIDERED.
- DELETED.
- DELETED.
- DELETED.
- ALL THE VALVES AROUND THE VESSEL, SHALL BE CLOSE TO THE VESSEL. DEAD LEG SHALL BE AVOIDED IN THE DRAIN LINES.
- SYMMETRICAL PIPING ARRANGEMENT SHALL BE PROVIDED FOR THE PSVs.
- MINIMUM RATING OF INSTRUMENTS SHALL BE 300#. FOR NOZZLE RATING REFER EQUIPMENT DATASHEET.
- EQUIPMENT IS FREE ISSUE ITEM BY ADOC AND ASSOCIATED INSTRUMENTS ARE IN SCOPE OF DETAILED ENGINEERING CONTRACTOR.
- DELETED.
- LG AND LT INSTRUMENT ISOLATION DBB, VENT AND DRAIN VALVE SCOPE DETAILS REFER INSTRUMENT HOOKUP DRAWING: PJ6-EXD-GEN-YZ0A-0002

LEGEND

 SCOPE CLOUD REVISION CLOUD

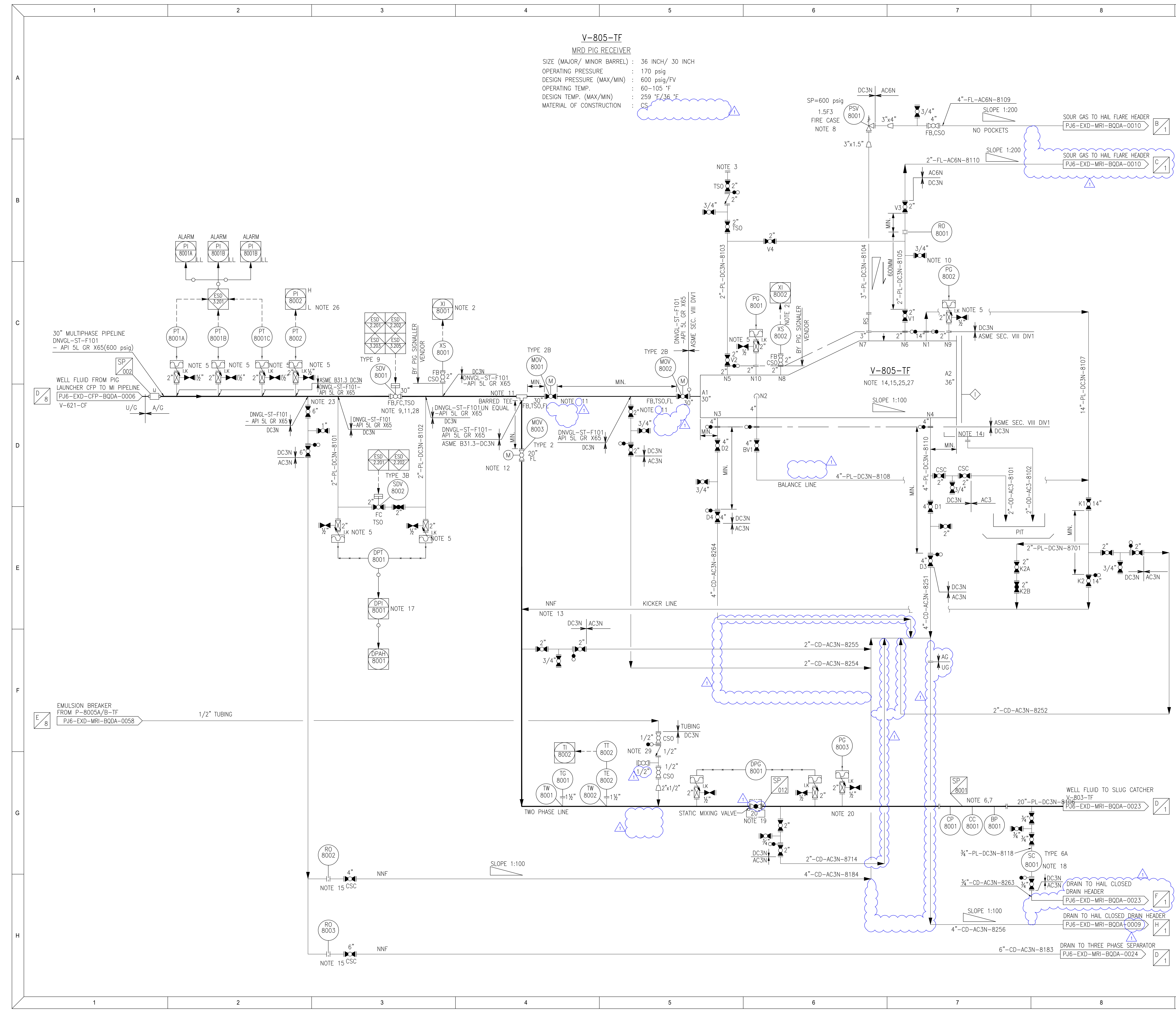
REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRI-BQDA-0005

THIS DRAWING HAS BEEN DEVELOPED FROM FEED DRAWING NO. PJ6-DF-MRI-BQDA-0001, REV 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	28/11/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
C	12/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	ZAS	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	ZAS	ASL	PKR
REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D
SCALE: NTS		LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002		
ENGINEERING CONSULTANT: REJLERS REJLERS TOWER 13TH FLOOR HAMDAN STREET, PO BOX 39317 ABU DHABI, UAE					
ENG CONSULTANT PROJECT NO: 5901055					
PMC : DORSCH HOLDING GmbH - ABU DHABI					
COMPANY: ABU DHABI OIL CO., LTD.					
PROJECT: DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)					
DRG. TITLE: PIPING AND INSTRUMENTATION DIAGRAM NEW MEA INLET SCRUBBER					
DWG. NO.	PJ6-EXD-MRI-BQDA-0021				REV. NO. 1 SHEET NO. 001

Filename : PJ6-EXD-MRI-BQDA-0021_1.DWG



V-805-TF
MRD PIG RECEIVER
SIZE (MAJOR/ MINOR BARREL) : 36 INCH/ 30 INCH
OPERATING PRESSURE : 170 psig
DESIGN PRESSURE (MAX/MIN) : 600 psig/FV
OPERATING TEMP. : 60-105 °F
DESIGN TEMP. (MAX/MIN) : 259 °F/36 °F
MATERIAL OF CONSTRUCTION : CS

NOTES

- ALL INSTRUMENT TAG NUMBERS ARE SUFFIXED WITH "TF" FOR MUBARRAZ ISLAND.
- INTRUSIVE TYPE PIG SIGNALLER.
- NITROGEN PURGING CONNECTION/WATER FLUSHING CONNECTION.
- QUICK OPENING END CLOSURE DOOR WILL BE EQUIPPED WITH A PRESSURE LOCKING DEVICE TO PREVENT OPENING OF DOORS WHEN THE PIG TRAP IS PRESSURIZED. A PRESSURE WARNING SAFETY INTERLOCK DEVICE SHALL ALSO BE PROVIDED AS PER AGES-SP-06-008.
- INSULATING KIT IS PROVIDED FOR DISSIMILAR METAL JOINTS.
- CORROSION COUPON & CORROSION PROBE WILL BE PROVIDED ON REMOVABLE SPOOL.
- CORROSION COUPON, CORROSION PROBE & BIO PROBE WILL BE RETRACTABLE UNDER PRESSURE. THE ACCESS FITTINGS ARE LOCATED AT THE HORIZONTAL LINE AT 6 O' CLOCK POSITION.
- WAREHOUSE SPARE PSV (TAG PSV-8008-TF) TO BE PROVIDED.
- CLOSING OF SDV (SDV-8001) WILL GENERATE AN ALARM IN DCS.
- PRESSURE INDICATOR (PG-8002) IS VISIBLE FROM PIG LAUNCHER DOOR. SCALE OF PRESSURE INDICATOR SHALL BE FROM 0 TO 15 psig, BUT PRESSURE RATED TO LINE PRESSURE.
- INTERNAL DIAMETER TO BE IDENTICAL TO PIPELINE INTERNAL DIAMETER.
- VALVE TO BE PROVIDED WITH INCHING FACILITY AND MECHANICAL STOP.
- AVOID DEAD LEG.
- VOLUME OF DRIP TRAY SHALL BE MINIMUM 5% OF BARREL VOLUME.
- LINE PROVIDED TO RECOVER OIL FROM PIPELINE IN CASE OF EMERGENCY.
- ALL DRAINS SHALL BE CONNECTED TO CLOSED DRAIN SYSTEM AND ALL VENTS SHALL BE CONNECTED TO FLARE.
- HIGH DIFFERENTIAL PRESSURE OF LESS THAN 100 psi SHALL GIVE PERMISSIVE TO OPEN SDV-8001.
- SAMPLE COLLECTION POINT SHALL BE ACCESSIBLE AT GROUND LEVEL.
- STATIC MIXING VALVE SHALL BE SUPPLIED BY PIG TRAP VENDOR.
- PRESSURE GAUGE SHALL BE READABLE FROM MIXING VALVE.
- MINIMUM RATING OF INSTRUMENTS SHALL BE 300#. FOR NOZZLE RATING
- REFER EQUIPMENT DATASHEET.
- DELETED
- SMALL BORE PIPING BRANCH UNDER ASME B 31.3 CODE WELDED TO DNV-ST-F101 PIPELINE WILL BE HYDROSTATICALLY TESTED JOINTLY WITH THE PIPELINE ACCORDING TO DNV CODE REQUIREMENT UP TO FIRST FLANGED JOINT AVAILABLE.
- EQUIPMENT IS FREE ISSUE ITEM BY ADOC AND ASSOCIATED INSTRUMENTS ARE IN SCOPE OF DETAILED ENGINEERING CONTRACTOR.
- PIPING TO PROVIDE ENOUGH SPACE AT THE END CLOSURE DOOR TO INSERT AND COLLECT PIG AT PIG LAUNCHER AND RECEIVER.
- PI-8002L IS SAFETY CRITICAL ALARM.
- PIG TRAPS SHALL BE PROVIDED WITH A PERMANENT JIB CRANE AND CHAIN PULLY BLOCK AT SITE.
- CONSIDER 1A VOLUME BOTTLE FOR SDV-8001.
- TWO DISSIMILAR TYPE CHECK VALVE SHALL BE USED AT INJECTION PUMP DISCHARGE AND AT THE INJECTION POINT.

LEGEND

- SCOPE CLOUD REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRI-BQDA-0005

THIS DRAWING HAS BEEN DEVELOPED FROM FEED DRAWING NO. PJ6-DF-MRI-BQDA-0002 REV 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	05/12/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
C	26/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	ZAS	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	ZAS	ASL	PKR

REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D
SCALE: NTS	LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002			

ENGINEERING CONSULTANT: **REJLERS** REJLERS TOWER 13TH FLOOR HAMDAN STREET, PO BOX 39317 ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

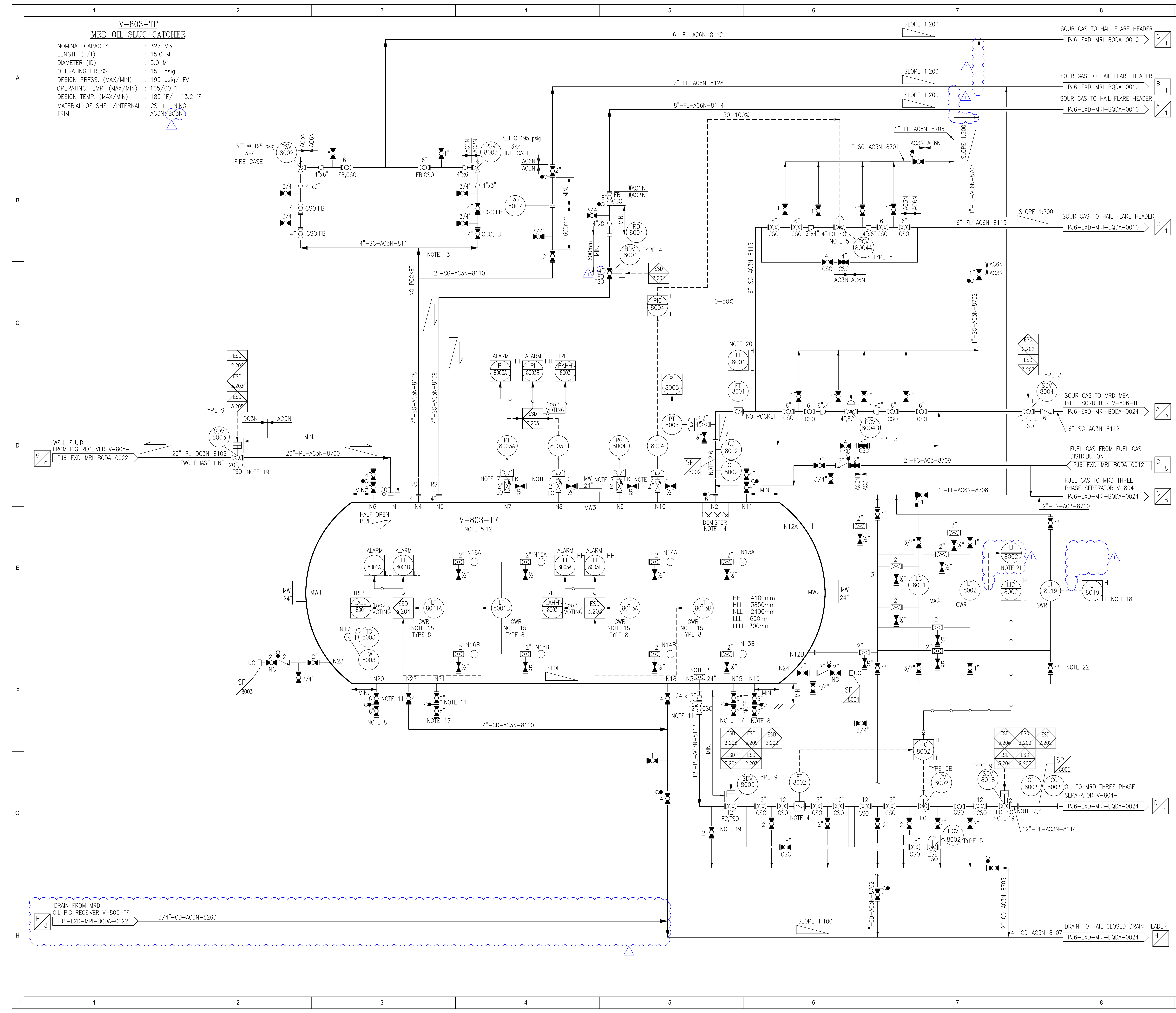
COMPANY: **ABU DHABI OIL CO., LTD.**

PROJECT: DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE: PIPING AND INSTRUMENTATION DIAGRAM PIG RECEIVER FOR CFP TO MI PIPELINE

DWG. NO.	PJ6-EXD-MRI-BQDA-0022	REV NO.	SHEET NO.
		1	001

Filename : PJ6-EXD-MRI-BQDA-0022.1.DWG



NOTES

- ALL INSTRUMENT TAG NUMBERS ARE SUFFIXED WITH "TF" FOR MUBARRAZ ISLAND.
- CORROSION COUPON & CORROSION PROBE WILL BE PROVIDED ON REMOVABLE SPOOL.
- LIQUID OUTLET NOZZLE TO BE PROVIDED WITH VORTEX BREAKER.
- ULTRASONIC FLOW METER WITH REDUCERS 100 UPSTREAM & 3D DOWNSTREAM.
- EQUIPMENT IS FREE ISSUE ITEM BY ADOC AND ASSOCIATED INSTRUMENTS ARE IN SCOPE OF DETAILED ENGINEERING CONTRACTOR.
- CORROSION COUPON AND CORROSION PROBE WILL BE RETRACTABLE UNDER PRESSURE. THE ACCESS FITTINGS ARE LOCATED AT THE HORIZONTAL LINE AT 6' 0" CLOCK POSITION.
- INSULATING KIT IS PROVIDED FOR DISSIMILAR METAL JOINTS.
- DRAIN POINT (LOW POINT) TO FACILITATE REMOVAL OF SLUDGE / SAND DURING MAINTENANCE.
- V-CONE TYPE FLOWMETER.
- MINIMUM RATING OF INSTRUMENTS SHALL BE 300#. FOR NOZZLE RATING REFER EQUIPMENT DATASHEET.
- ALL THE VALVES AROUND THE VESSEL, SHALL BE CLOSE TO THE VESSEL. DEAD LEG SHALL BE AVOIDED IN THE DRAIN LINES.
- GASKET FOR ALL THE VALVES ASSOCIATED WITH THE VESSEL SHALL BE SUITABLE FOR STEAM.
- SYMMETRICAL PIPING ARRANGEMENT SHALL BE CONSIDERED FOR THE PSVs.
- MESH PAD TYPE DEMISTER SHALL BE PROVIDED AT THE GAS OUTLET.
- LEVEL INSTRUMENT SHALL BE SUITABLE FOR EMULSION.
- ALL DRAINS SHALL BE CONNECTED TO CLOSED DRAIN SYSTEM AND ALL VENTS SHALL BE CONNECTED TO FLARE.
- PROVISION FOR JETTING WATER CONNECTION.
- LI-8019L IS SAFETY CRITICAL ALARM.
- CONSIDER IA VOLUME BOTTLE FOR SDV-8003, SDV-8005 AND SDV-8018.
- LI-8001(F) IS SAFETY CRITICAL ALARM.
- LI-8002 READABLE FROM LCV-8002 BYPASS GLOBE VALVE.
- LG AND LT INSTRUMENT ISOLATION DBB, VENT AND DRAIN VALVE SCOPE DETAILS REFER INSTRUMENT HOOKUP DRAWING.PJ6-EXD-GEN-YZ0A-0002

LEGEND

 SCOPE CLOUD REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRI-BQDA-0005

THIS DRAWING HAS BEEN DEVELOPED FROM FEED DRAWING NO. PJ6-DF-MRI-BQDA-0003 REV 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D
1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	05/12/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
C	26/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	ZAS	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	ZAS	ASL	PKR

SCALE:	LOCATION:	COMPANY PROJECT NO.
NTS	MUBARRAZ ISLAND	PJ-2025-CR-002

ENGINEERING CONSULTANT:

REJLERS

REJLERS TOWER 13TH FLOOR
HAMDAN STREET, PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO:5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

COMPANY:

ABU DHABI OIL CO., LTD.

PROJECT:
DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE:
PIPING AND INSTRUMENTATION DIAGRAM
NOW OIL SLUG CATCHER (V-308-TF)

DWG. NO.	PJ6-EXD-MRI-BQDA-0023	REV/NO.	SHEET NO.
		1	001

Filename : PJ6-EXD-MRI-BQDA-0023.1.DWG

SHT. SIZE: A1

V-804-TF

MRD THREE PHASE SEPARATOR

NOMINAL CAPACITY : 120 m³
LENGTH (T/T) : 12000 mm
DIAMETER (ID) : 3400 mm
OPERATING PRESS. : 110 psig
DESIGN PRESS. (MAX/MIN) : 185 psig/ FV
OPERATING TEMP.(MAX/MIN) : 105/60 °F
DESIGN TEMP. (MAX/MIN) : 185 °F/25 °F
MATERIAL OF SHELL/INTERNAL: CS + LINING
TRIM : AC3N

SOUR GAS FROM MRD OIL
SLUG CATCHER V-803-TF
PJ6-DF-MRI-BQDA-0003

6"-SG-AC3N-8112

10"-SG-AC3N-8122

SOUR GAS TO MRD MEA
INLET SCRUBBER V-806-TF
PJ6-EXD-MRI-BQDA-0014

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

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PJ6-EXD-MRI-BQDA-0010

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SOUR GAS TO HAIL FLARE HEADER
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SOUR GAS TO HAIL FLARE HEADER
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SOUR GAS TO HAIL FLARE HEADER
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SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
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SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
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SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

SOUR GAS TO HAIL FLARE HEADER
PJ6-EXD-MRI-BQDA-0010

NOTES

- DELETED.
- SAMPLE COLLECTION POINT SHALL BE ACCESSIBLE AT GROUND LEVEL.
- NO POCKET. MINIMUM BEND.
- HC LIQUID & WATER OUTLET NOZZLES TO BE PROVIDED WITH VORTEX BREAKER.
- INTERFACE LOW LEVEL TRIP TO AVOID OIL GAS BLOW-BY CASE.
- SEPARATOR INTERNALS: HALF OPEN PIPE (INLET DEVICE) SUBMERGED WEIR, MIST ELIMINATOR, VORTEX BREAKERS AND PERFORATED PLATE.
- LI-8005 READABLE FROM LCV-8005 BYPASS GLOBE VALVE.
- MAGNETIC TYPE FLOWMETER.
- DRAIN POINT TO FACILITATE REMOVAL OF SLUDGE/SAND DURING MAINTENANCE.
- CORROSION COUPON & CORROSION PROBE WILL BE PROVIDED ON REMOVABLE SPOOL.
- CORROSION COUPON AND CORROSION PROBE WILL BE RETRACTABLE UNDER PRESSURE. THE ACCESS FITTINGS ARE LOCATED AT THE HORIZONTAL LINE AT 6 O' CLOCK POSITION.
- ULTRASONIC FLOWMETER.
- ALL THE VALVES AROUND THE VESSEL, SHALL BE CLOSE TO THE VESSEL. DEAD LEG SHALL BE AVOIDED IN THE DRAIN LINES.
- GASKET FOR ALL THE VALVES ASSOCIATED WITH THE VESSEL SHALL BE SUITABLE FOR STEAM.
- SYMMETRICAL PIPING ARRANGEMENT SHALL BE CONSIDERED FOR THE PSVS.
- LG AND LT INSTRUMENT ISOLATION DBB, VENT AND DRAIN VALVE SCOPE DETAILS REFER INSTRUMENT HOOKUP DRAWING.PJ6-EXD-GEN-Y20A-0002
- PROVISION FOR JETTING WATER CONNECTION.
- MINIMUM STRAIGHT LENGTH REQUIRED AT UPSTREAM AND DOWNSTREAM TO BE 10D AND 5D RESPECTIVELY.
- ALL DRAINS SHALL BE CONNECTED TO CLOSED DRAIN SYSTEM AND ALL VENTS SHALL BE CONNECTED TO FLARE.
- MESH PAD TYPE DEMISTER SHALL BE PROVIDED AT THE GAS OUTLET.
- LI-8008 READABLE FROM LCV-8008 BYPASS GLOBE VALVE.
- MINIMUM RATING OF INSTRUMENTS SHALL BE 300#. FOR NOZZLE RATING REFER EQUIPMENT DATASHEET.
- NOZZLE SHALL BE KEPT AT THE CENTER OF THE VESSEL.
- EQUIPMENT IS FREE ISSUE ITEM BY ADCC AND ASSOCIATED INSTRUMENTS ARE IN SCOPE OF DETAILED ENGINEERING CONTRACTOR.
- FLOW TO MRD PRODUCED WATER TREATMENT PACKAGE SHALL BE LIMITED TO 81,440 BWPD.
- DELETED.
- 5 NO OF SAMPLE THIEF PROVISION TO BE PROVIDED IN THE MRD THREE PHASE SEPERATOR.
- FI-8003H IS SAFETY CRITICAL ALARM.

LEGEND

- SCOPE CLOUD
- REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRI-BQDA-0005

THIS DRAWING HAS BEEN DEVELOPED FROM FEED DRAWING NO. PJ6-DF-MRI-BQDA-0004 REV 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D
1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	28/11/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
C	12/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	SRK	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	SRK	ASL	PKR
SCALE: NTS		LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002		

ENGINEERING CONSULTANT: REJLERS
REJLERS TOWER 13TH FLOOR
HAMDAN STREET,PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO:5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

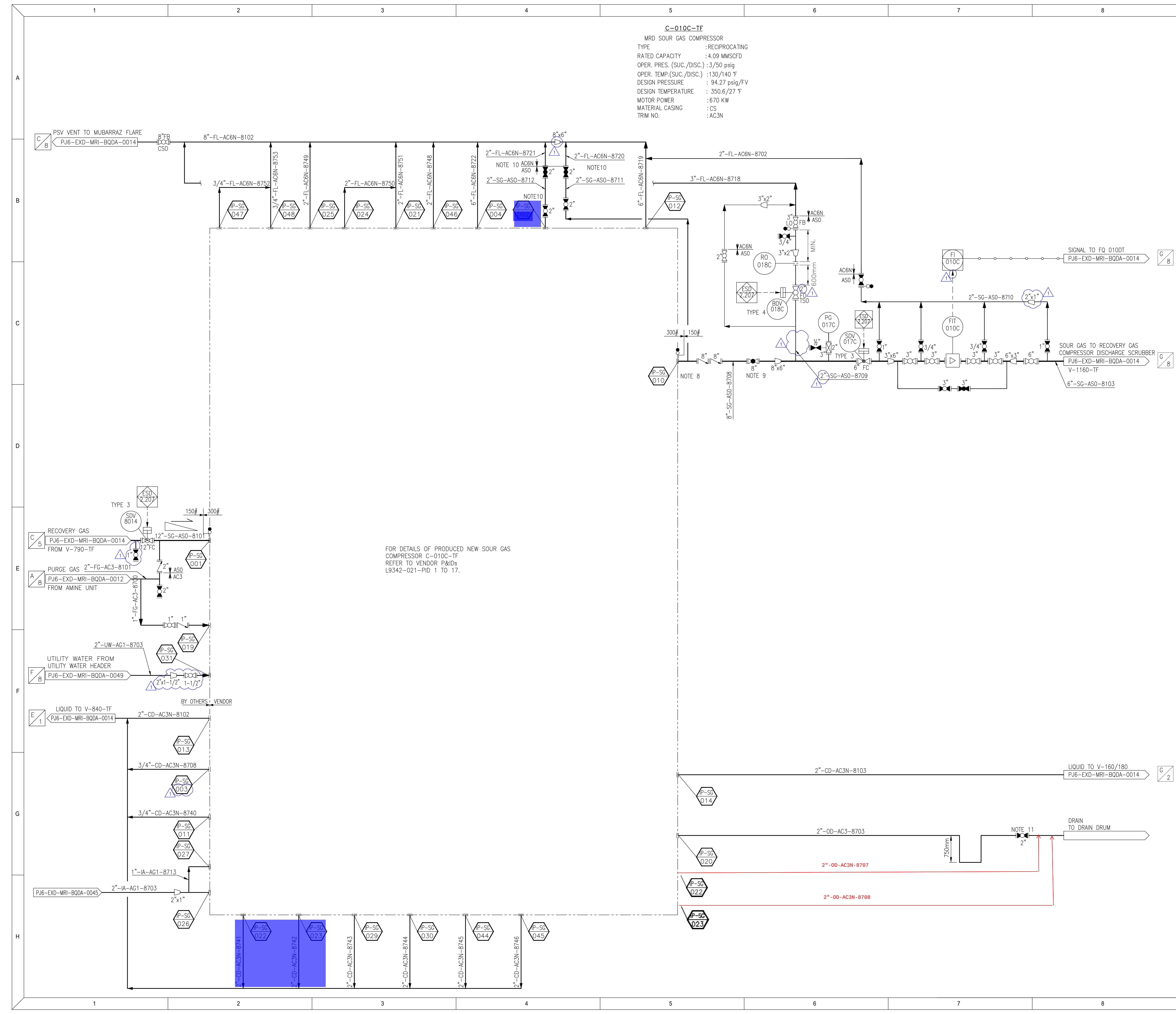
COMPANY: **ABU DHABI OIL CO.,LTD.**

PROJECT: DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE: PIPING AND INSTRUMENTATION DIAGRAM
NEW MUBARRAZ THREE PHASE SEPARATOR

DWG. NO.	PJ6-EXD-MRI-BQDA-0024	REV NO.	SHEET NO.
		1	001

Filename : PJ6-EXD-MRI-BQDA-0024_1.DWG



NOTES

1. ALL INSTRUMENT TAG NUMBERS SHALL BE SUFFIXED BY SITE SYMBOL CODE TF.

2. DELETED.

3. DELETED.

4. ALL DRAINS SHALL BE CONNECTED TO CLOSED DRAIN SYSTEM AND ALL VENTS SHALL BE CONNECTED TO FLARE.

5. INSULATING KIT IS PROVIDED FOR DISSIMILAR METAL JOINTS.

6. MINIMUM RATING OF INSTRUMENTS SHALL BE 300#.

7. INTERCONNECTING PIPING BETWEEN SKID AND EQUIPMENT BY PC CONTRACTOR.

8. TWO NRVs PROVIDED IN SERIES AT THE COMPRESSOR OUTLET SCRUBBER (V-105E) DISCHARGE LINE SHALL BE DISSIMILAR TYPE.

9. 1 NO OF 8" NRV IS LOOSE SUPPLY BY COMPRESSOR VENDOR. 2ND DISSIMILAR NRV BY PC SCOPE. 8" ISOLATION VALVE LOOSE SUPPLY BY COMPRESSOR VENDOR.

10. LOOSE SUPPLY BY COMPRESSOR VENDOR.

11. DISTANCE PIECE DRAIN TO BE ROUTED TO DRAIN DRUM.

LEGEND

SCOPE CLOUD

VENDOR TIE-IN POINT

REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

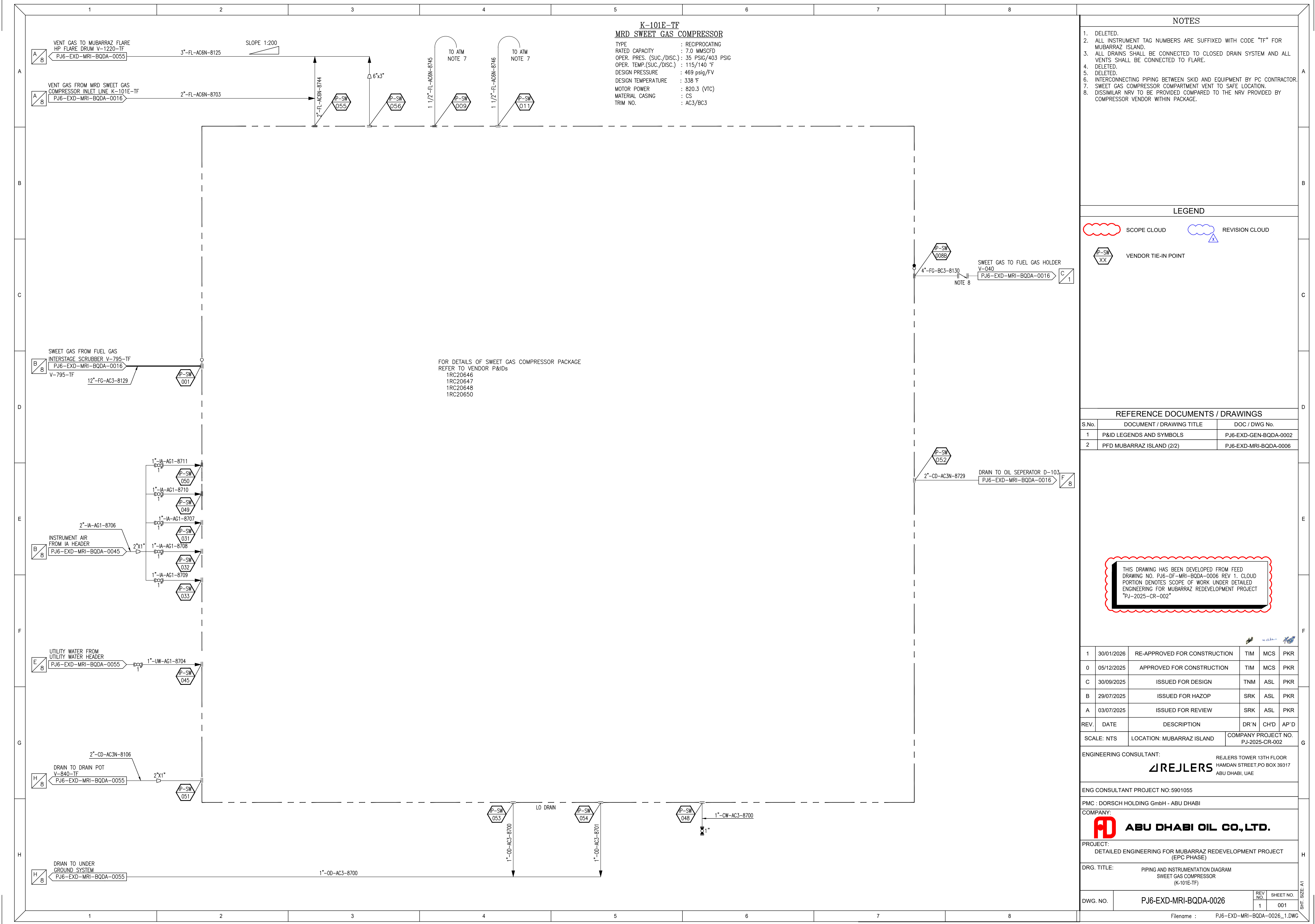
S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-MRF-BQDA-0050
2	PFD MUBARRAZ ISLAND (1/2)	PJ6-EXD-MRI-BQDA-0005

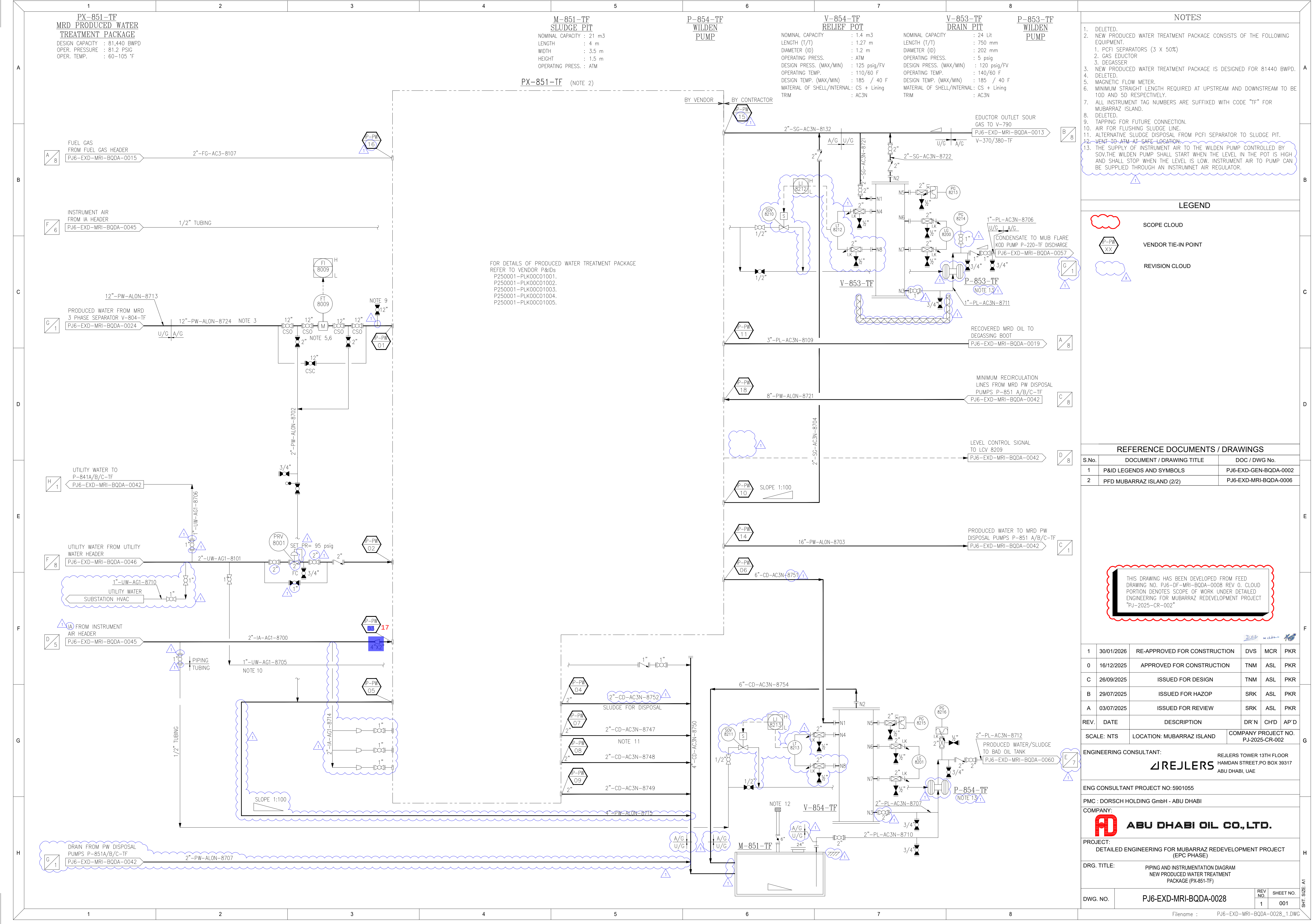
THIS DRAWING HAS BEEN DEVELOPED FROM FEED DRAWING NO. PJ6-DF-MRI-BQDA-0005 REV 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	05/12/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
C	12/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	SRK	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	SRK	ASL	PKR
REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D
SCALE: NTS		LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002		
ENGINEERING CONSULTANT:		REJLERS REJLERS TOWER 13TH FLOOR HAMDAN STREET,PO BOX 39317 ABU DHABI, UAE			
ENG CONSULTANT PROJECT NO:5901055					
PMC : DORSCH HOLDING GmbH - ABU DHABI					
COMPANY: ABU DHABI OIL CO.,LTD.					
PROJECT: DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)					
DRG. TITLE: PIPING AND INSTRUMENTATION DIAGRAM NEW SOUR GAS COMPRESSOR (C-010C-TF)					
DWG. NO.	PJ6-EXD-MRI-BQDA-0025				REV NO. 1 SHEET NO. 001

Filename : PJ6-EXD-MRI-BQDA-0025_1.DWG

SHT. SIZE: A1





- NOTES
1. DELETED.

2. NEW PRODUCED WATER TREATMENT PACKAGE CONSISTS OF THE FOLLOWING EQUIPMENT.

1. PCFI SEPARATORS (3 X 50%)

2. GAS EDUCTOR

3. DEGASSER

3. NEW PRODUCED WATER TREATMENT PACKAGE IS DESIGNED FOR 81440 BWPd.

4. DELETED.

5. MAGNETIC FLOW METER.

6. MINIMUM STRAIGHT LENGTH REQUIRED AT UPSTREAM AND DOWNSTREAM TO BE 10D AND 5D RESPECTIVELY.

7. ALL INSTRUMENT TAG NUMBERS ARE SUFFIXED WITH CODE "TF" FOR MUBARRAZ ISLAND.

8. DELETED.

9. TAPPING FOR FUTURE CONNECTION.

10. AIR FOR FLUSHING SLUDGE LINE.

11. ALTERNATIVE SLUDGE DISPOSAL FROM PCFI SEPARATOR TO SLUDGE PIT.

12. VENT TO ATM AT SAFE LOCATION.

13. THE SUPPLY OF INSTRUMENT AIR TO THE WILDEN PUMP CONTROLLED BY SOV.THE WILDEN PUMP SHALL START WHEN THE LEVEL IN THE POT IS HIGH AND SHALL STOP WHEN THE LEVEL IS LOW. INSTRUMENT AIR TO PUMP CAN BE SUPPLIED THROUGH AN INSTRUMNET AIR REGULATOR.

- LEGEND
- SCOPE CLOUD

VENDOR TIE-IN POINT

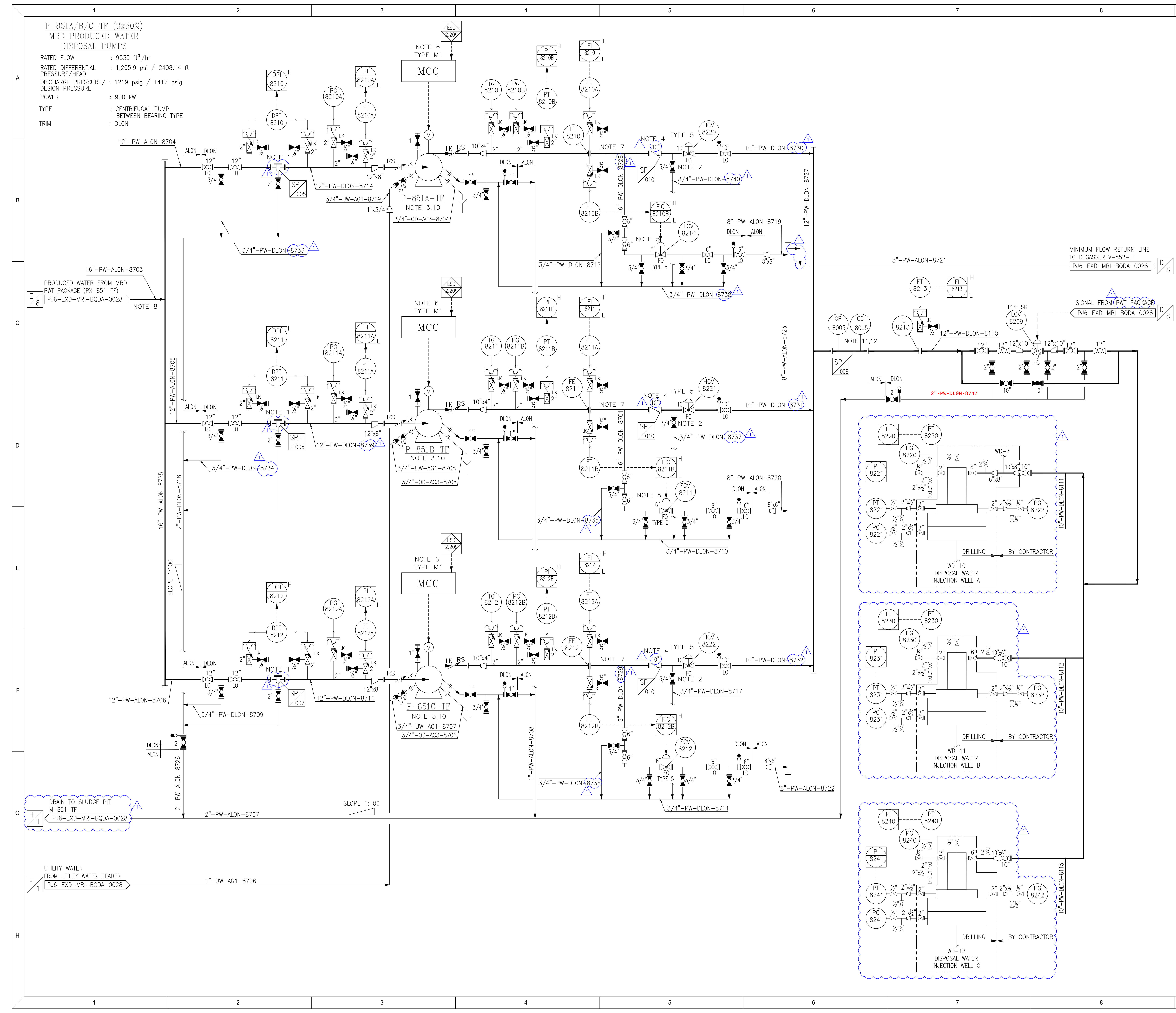
REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (2/2)	PJ6-EXD-MRI-BQDA-0006

THIS DRAWING HAS BEEN DEVELOPED FROM FEED DRAWING NO. PJ6-DF-MRI-BQDA-0008 REV 0. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCR	PKR
0	16/12/2025	APPROVED FOR CONSTRUCTION	TNM	ASL	PKR
C	26/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	SRK	ASL	PKR
A	03/07/2025	ISSUED FOR REVIEW	SRK	ASL	PKR
REV.	DATE	DESCRIPTION	DR 'N	CH'D	AP'D
SCALE:	NTS	LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002		
ENGINEERING CONSULTANT:			REJLERS TOWER 13TH FLOOR HAMDAN STREET,PO BOX 39317 ABU DHABI, UAE		
ENG CONSULTANT PROJECT NO: 5901055					
PMC : DORSCH HOLDING GmbH - ABU DHABI					
COMPANY:			ABU DHABI OIL CO.,LTD.		
PROJECT:			DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)		
DRG. TITLE:			PIPING AND INSTRUMENTATION DIAGRAM NEW PRODUCED WATER TREATMENT PACKAGE (PX-851-TF)		
DWG. NO.	PJ6-EXD-MRI-BQDA-0028		REV NO.	1	SHEET NO. 001



NOTES

1. PERMANENT STRAINER (BUCKET TYPE).

2. VALVE TO OPEN AND CLOSE SLOWLY TO PREVENT HAMMER EFFECT. VALVE TO OPEN WITH ACTIVATION OF ASSOCIATED PUMP.

3. PUMP SUPPLIED WITH API 610 PLAN 53B MECHANICAL SEAL.

4. NON SLAM TYPE CHECK VALVE (WITH 5D UPSTREAM & 3D DOWNSTREAM).

5. HIGH INTEGRITY ANTI CAVITY TRIM.

6. PUMP CONFIGURATION: 3 X 50% (2 DUTY + 1 STANDBY). STANDBY PUMP SHALL START AUTOMATICALLY IN THE EVENT OF FAILURE OF EITHER DUTY PUMP.

7. ORIFICE TAPING TYPE SHALL BE FLANGE TAPS. THE TAPPINGS ON ORIFICE FLANGES FOR MONITORING/CONTROL AND ESD/TRIP APPLICATIONS SHALL NOT BE ON THE SAME POINTS. FOR TAPPING ORIENTATION, REFER TO INSTRUMENT HOOK-UP DRAWING PJ6-EXD-GEN-Y20A-0002.

8. PUMP SUCTION AND DISCHARGE TO BE SYMMETRICAL ARRANGEMENT.

9. ALL PUMP CONTROL WILL BE FROM VENDOR PACKAGE PLC.

10. PRODUCED WATER DISPOSAL PUMP AND RELATED INSTRUMENTS WILL BE SUPPLIED BY PUMP VENDOR AND ALL OTHER INSTRUMENT ARE UNDER PC CONTRACTOR.

11. CORROSION COUPON & CORROSION PROBE WILL BE PROVIDED ON REMOVABLE SPOOL.

12. CORROSION COUPON AND CORROSION PROBE WILL BE RETRACTABLE UNDER PRESSURE. THE ACCESS FITTINGS ARE LOCATED AT THE HORIZONTAL LINE AT 6 O' CLOCK POSITION.

LEGEND

REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (2/2)	PJ6-EXD-MRI-BQDA-0006

1

30/01/2026

RE-APPROVED FOR CONSTRUCTION

DVS

MCS

PKR

0

16/12/2025

APPROVED FOR CONSTRUCTION

TIM

MCS

PKR

C

12/09/2025

ISSUED FOR DESIGN

TNM

ASL

PKR

B

29/07/2025

ISSUED FOR HAZOP

ZAS

ASL

PKR

A

03/07/2025

ISSUED FOR REVIEW

ZAS

ASL

PKR

REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D
SCALE: NTS	LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002			

ENGINEERING CONSULTANT:

REJLERS

REJLERS TOWER 13TH FLOOR
HAMDAN STREET, PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

COMPANY:

ABU DHABI OIL CO., LTD.

PROJECT:

DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

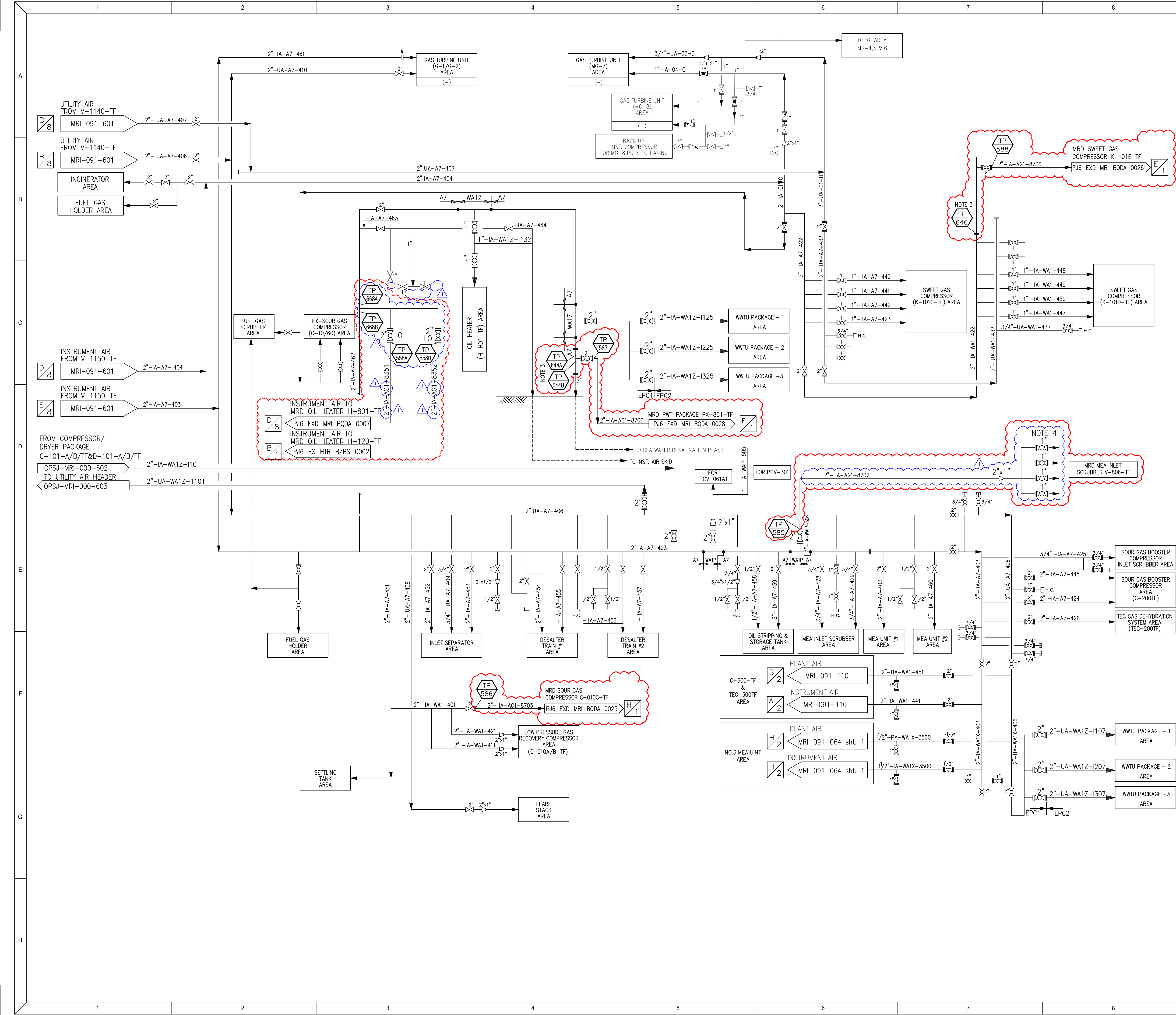
DRG. TITLE:

PIPING AND INSTRUMENTATION DIAGRAM
NEW PRODUCED WATER DISPOSAL
PUMPS (P-851 A/B/C-TF)

DWG. NO.	PJ6-EXD-MRI-BQDA-0042	REV NO.	SHEET NO.
		1	001

Filename : PJ6-EXD-MRI-BQDA-0042_1.DWG

SHT. SIZE: A1



NOTES

1. AS BUILT INFORMATION

2. DELETED

3. EARLY TIE-IN (TP 644 A/B, 668A/B) SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN

4. 1" FLANGE WITH FEMALE NPT TO BE PROVIDED

LEGEND

SCOPE CLOUD

SCOPE CLOUD OF "DETAIL ENGINEERING FOR INSTALLATION OF NEW CRUDE OIL HEATERS (H-120-TF & H-801-TF) AT MUBARRAZ ISLAND" PROJECT.

TP XX

TIE-IN POINT

REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	P&ID INSTRUMENT AND PLANT AIR SUPPLY SYSTEM (DEMOLITION)	PJ6-EXD-MRI-BQDA-0047

THIS DRAWING HAS BEEN DEVELOPED FROM DRAWING NO. 6NCP-MRI-000-003 Rev 1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002".

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	14/11/2025	APPROVED FOR CONSTRUCTION	TNM	ASL	PKR
C	12/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	SKA	ASL	PKR
A	02/07/2025	ISSUED FOR REVIEW	SKA	ASL	PKR

REV.	DATE	DESCRIPTION	DR 'N	CH'D	AP'D
SCALE:	NTS	LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002		

ENGINEERING CONSULTANT:

IREJLERS

REJULERS TOWER 13TH FLOOR
HAMDAN STREET, PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC : DORSCH HOLDING GmbH - ABU DHABI

COMPANY:

ABU DHABI OIL CO., LTD.

PROJECT:

DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

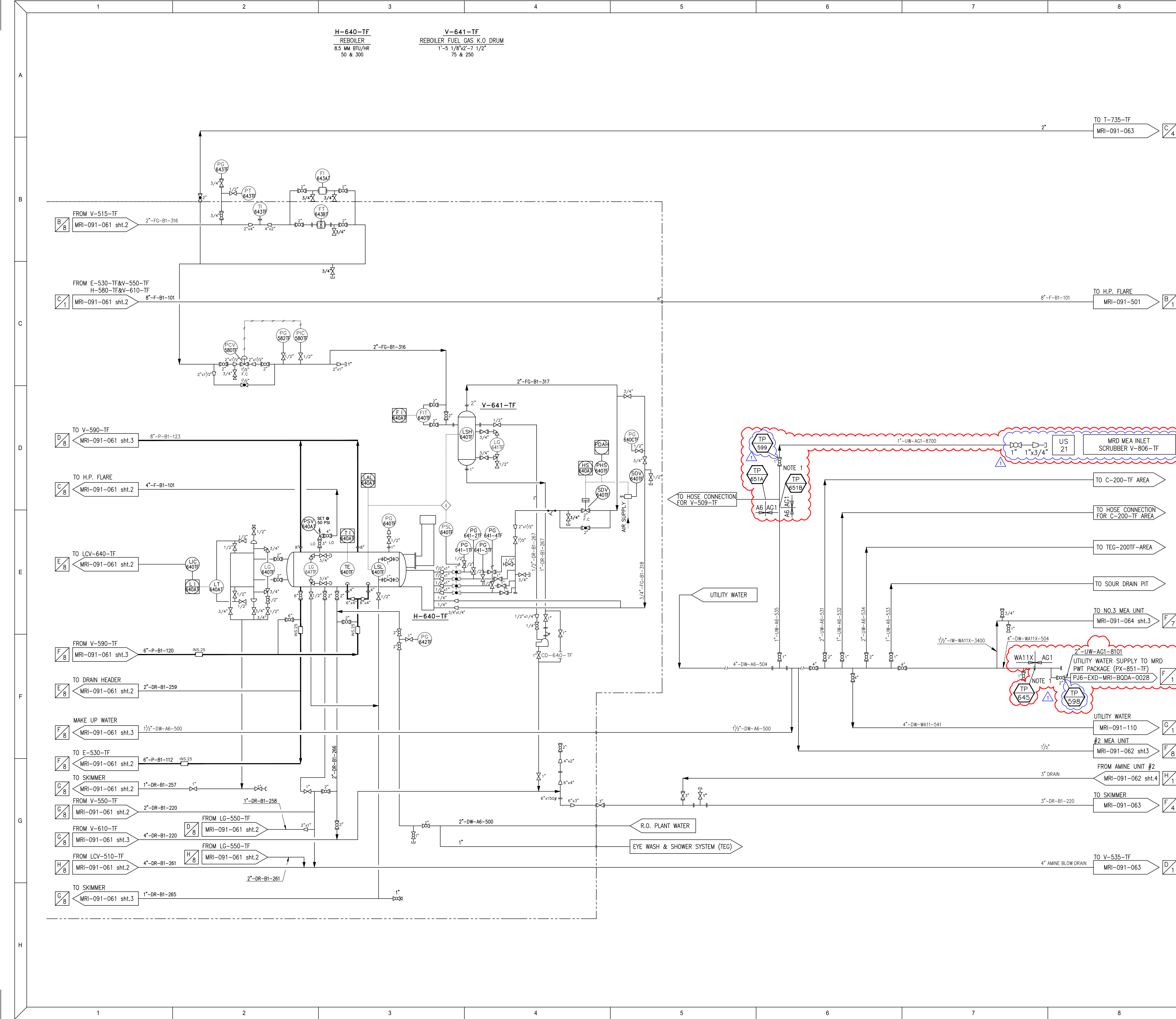
DRG. TITLE:

PIPING AND INSTRUMENTATION DIAGRAM
INSTRUMENT AND PLANT AIR SUPPLY SYSTEM

DWG. NO.	PJ6-EXD-MRI-BQDA-0045	REV NO.	1	SHEET NO.	001
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Filename : PJ6-EXD-MRI-BQDA-0045_1.DWG

SHEET SIZE A1



NOTES

1. EARLY TIE-IN (TP-645,651 A/B) SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.

LEGEND

SCOPE CLOUD

TIE-IN POINT

REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	P&ID INSTRUMENT AND PLANT AIR SUPPLY SYSTEM (DEMOLITION)	PJ6-EXD-MRI-BQDA-0048

THIS DRAWING HAS BEEN DEVELOPED FROM DRAWING NO. MRI-091-061 SHT 4 Rev 12. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002".

REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D
1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCR	PKR
0	14/10/2025	APPROVED FOR CONSTRUCTION	TNM	ASL	PKR
C	12/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
B	29/07/2025	ISSUED FOR HAZOP	SKA	ASL	PKR
A	02/07/2025	ISSUED FOR REVIEW	SKA	ASL	PKR

SCALE:	LOCATION:	COMPANY PROJECT NO.
NTS	MUBARRAZ ISLAND	PJ-2025-CR-002

ENGINEERING CONSULTANT:

REJLERS

REJLERS TOWER 13TH FLOOR
HAMDAN STREET, PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO:

5901055

PMC:

DORSCH HOLDING GmbH - ABU DHABI

COMPANY:

AD

ABU DHABI OIL CO.,LTD.

PROJECT:

DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

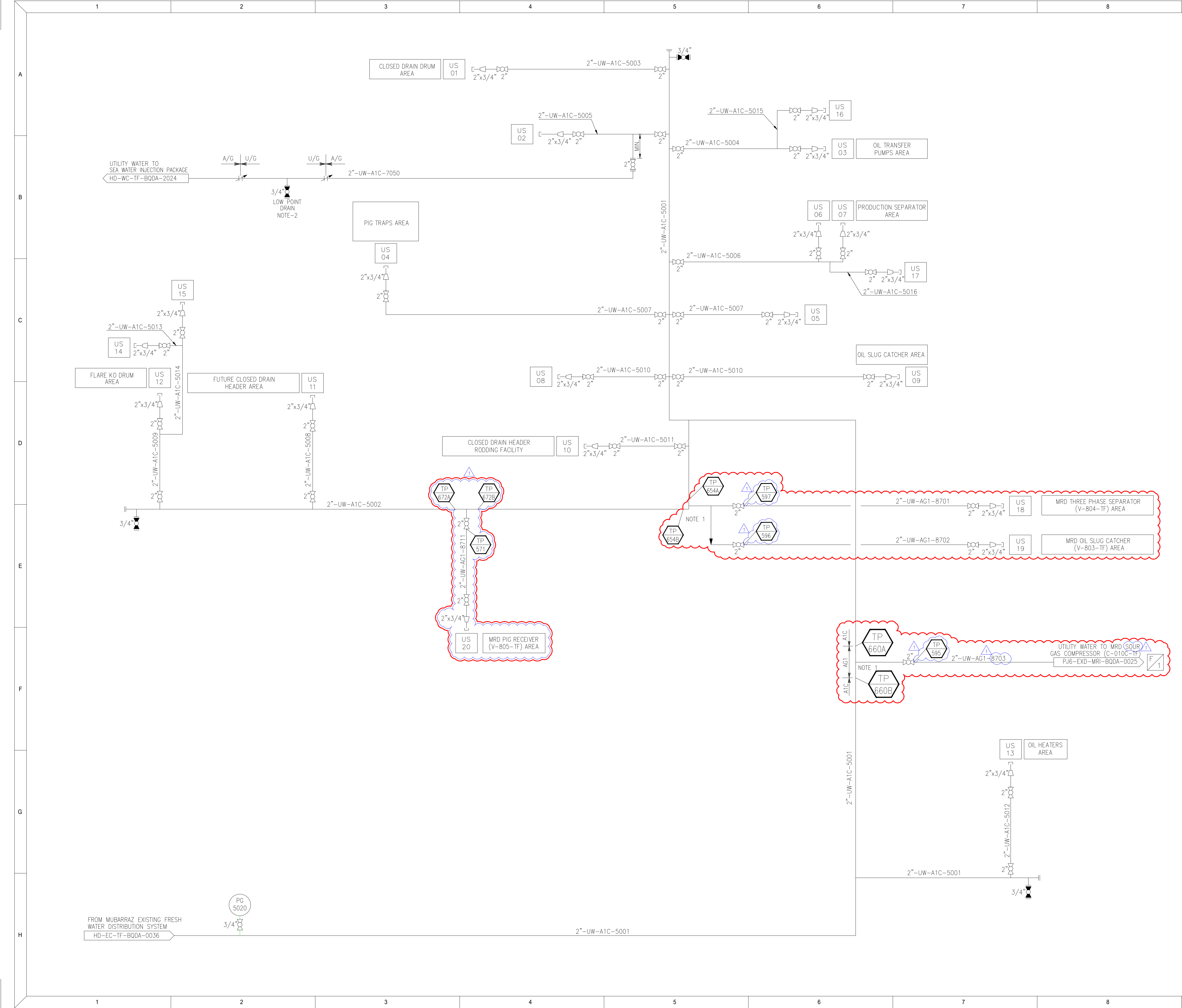
DRG. TITLE:

PIPING AND INSTRUMENTATION DIAGRAM
AMINE UNIT #1 (4 OF 4)

DWG. NO.	PJ6-EXD-MRI-BQDA-0046	REV NO.	SHEET NO.
		1	001

Filename : PJ6-EXD-MRI-BQDA-0046_1.DWG

SHEET SIZE: A1



NOTES

1. EARLY TIE-IN (TP-654A/B, 660A/B) SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.

2. TO BE LOCATED INSIDE PIT.

LEGEND

SCOPE CLOUD

TIE-IN POINT

REVISION CLOUD

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	P&ID FRESH WATER (DEMOLITION)	PJ6-EXD-MRI-BQDA-0050

THIS DRAWING HAS BEEN DEVELOPED FROM HAIL OIL FIELD WATER INJECTION PROJECT (WIP) DRAWING NO. HD-WC-TF-BQDA-2318 SHT 2 OF 2 REV-1. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

DESIGNED:

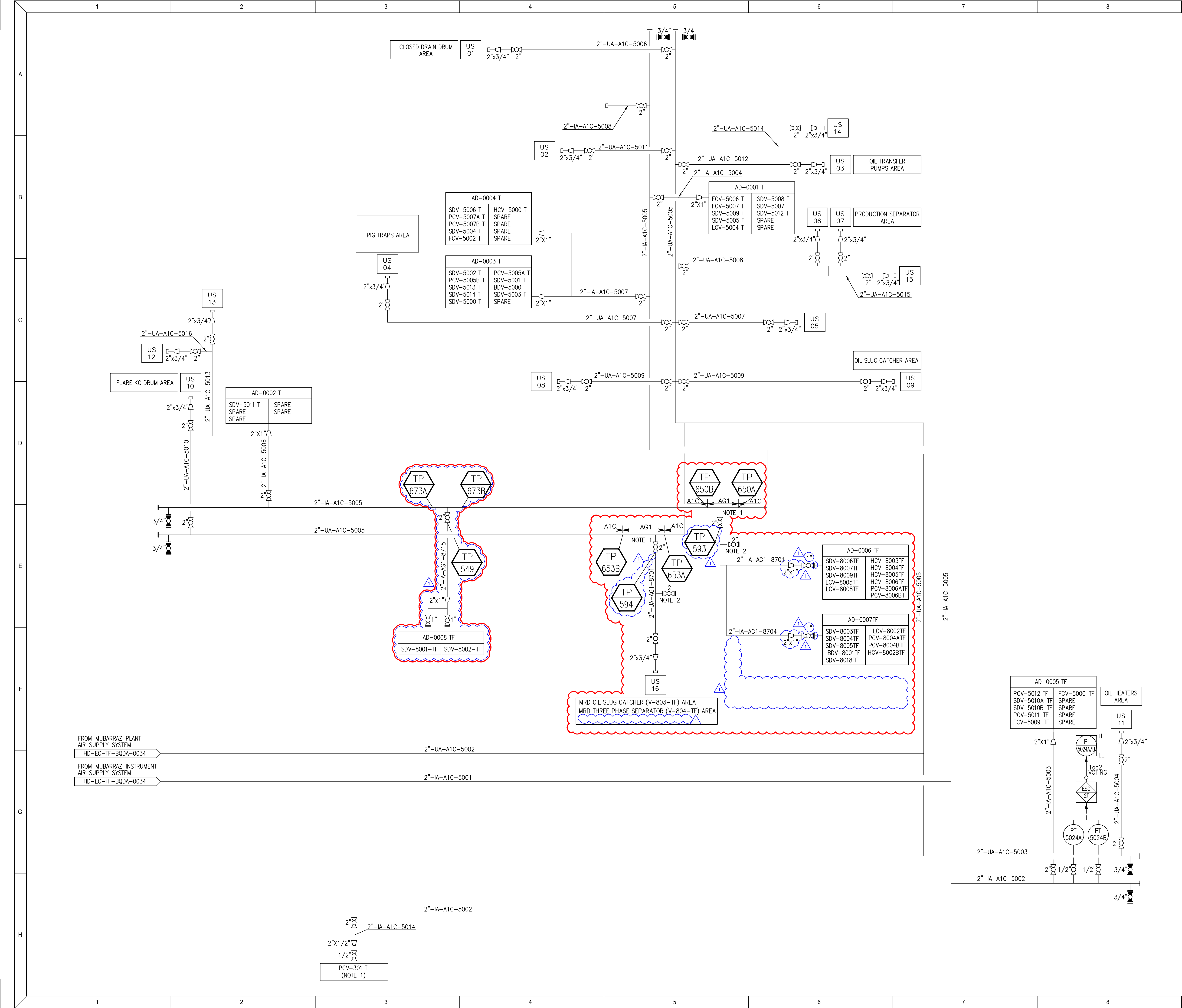
CHECKED:

APPROVED:

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	ASL	PKR
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B	12/09/2025	ISSUED FOR DESIGN	ZAS	ASL	PKR
A	31/07/2025	ISSUED FOR REVIEW	ZAS	ASL	PKR
REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D
SCALE:	NTS	LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002		
ENGINEERING CONSULTANT:			REJILERS TOWER 13TH FLOOR HAMDAN STREET, PO BOX 39317 ABU DHABI, UAE		
ENG CONSULTANT PROJECT NO: 5901055					
PMC : DORSCH HOLDING GmbH - ABU DHABI					
COMPANY: ABU DHABI OIL CO.,LTD.					
PROJECT: DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)					
DRG. TITLE: PIPING AND INSTRUMENTATION DIAGRAM FRESH WATER					
DWG. NO.	PJ6-EXD-MRI-BQDA-0049			REV NO.	SHEET NO.
				1	001

Filename : PJ6-EXD-MRI-BQDA-0049_1.DWG

SHT SIZE: A1



NOTES

1. EARLY TIE-IN TP-650A/B,TP-653A/B SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.

2. VALVE SHALL BE USED FOR FUTURE CONNECTION.

LEGEND

SCOPE CLOUD

TP
XX

TIE-IN POINT

REVISION CLOUD

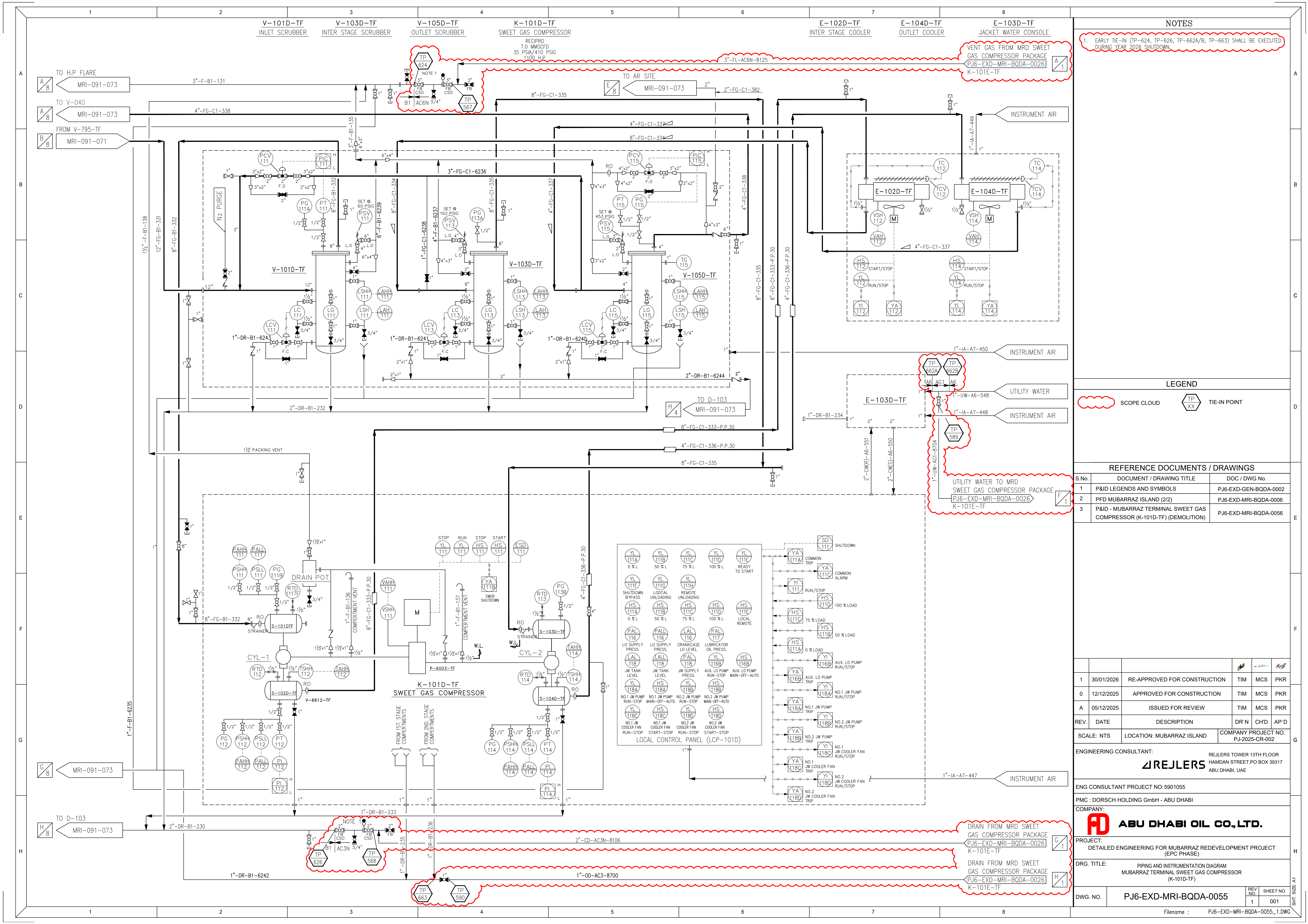
REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	P&ID INSTRUMENT AND PLANT AIR DISTRIBUTION (DEMOLITION)	PJ6-EXD-MRI-BQDA-0054

THIS DRAWING HAS BEEN DEVELOPED FROM PROJECT NO : 8598 DRAWING NO. HD-EC-TF-BQDA-0027 SHT 1 OF 1 Rev 4. CLOUD PORTION DENOTES SCOPE OF WORK UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	14/11/2025	APPROVED FOR CONSTRUCTION	TNM	ASL	PKR
B	12/09/2025	ISSUED FOR DESIGN	TNM	ASL	PKR
A	26/08/2025	ISSUED FOR REVIEW	TNM	ASL	PKR
REV.	DATE	DESCRIPTION	DR'N	CHD	AP'D
SCALE:	NTS	LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002		
ENGINEERING CONSULTANT:			REJILERS TOWER 13TH FLOOR REJILERS HAMDAN STREET,PO BOX 39317 ABU DHABI, UAE		
ENG CONSULTANT PROJECT NO: 5901055					
PMC : DORSCH HOLDING GmbH - ABU DHABI					
COMPANY: ABU DHABI OIL CO.,LTD.					
PROJECT: DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)					
DRG. TITLE: PIPING AND INSTRUMENTATION DIAGRAM INSTRUMENT AND PLANT AIR DISTRIBUTION					
DWG. NO.	PJ6-EXD-MRI-BQDA-0053				REV NO. 1 SHEET NO. 001

Filename : PJ6-EXD-MRI-BQDA-0053_1.DWG



NOTES

1. EARLY TIE-IN (TP-624, TP-626, TP-662A/B, TP-663) SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.

LEGEND

SCOPE CLOUD

TIE-IN POINT

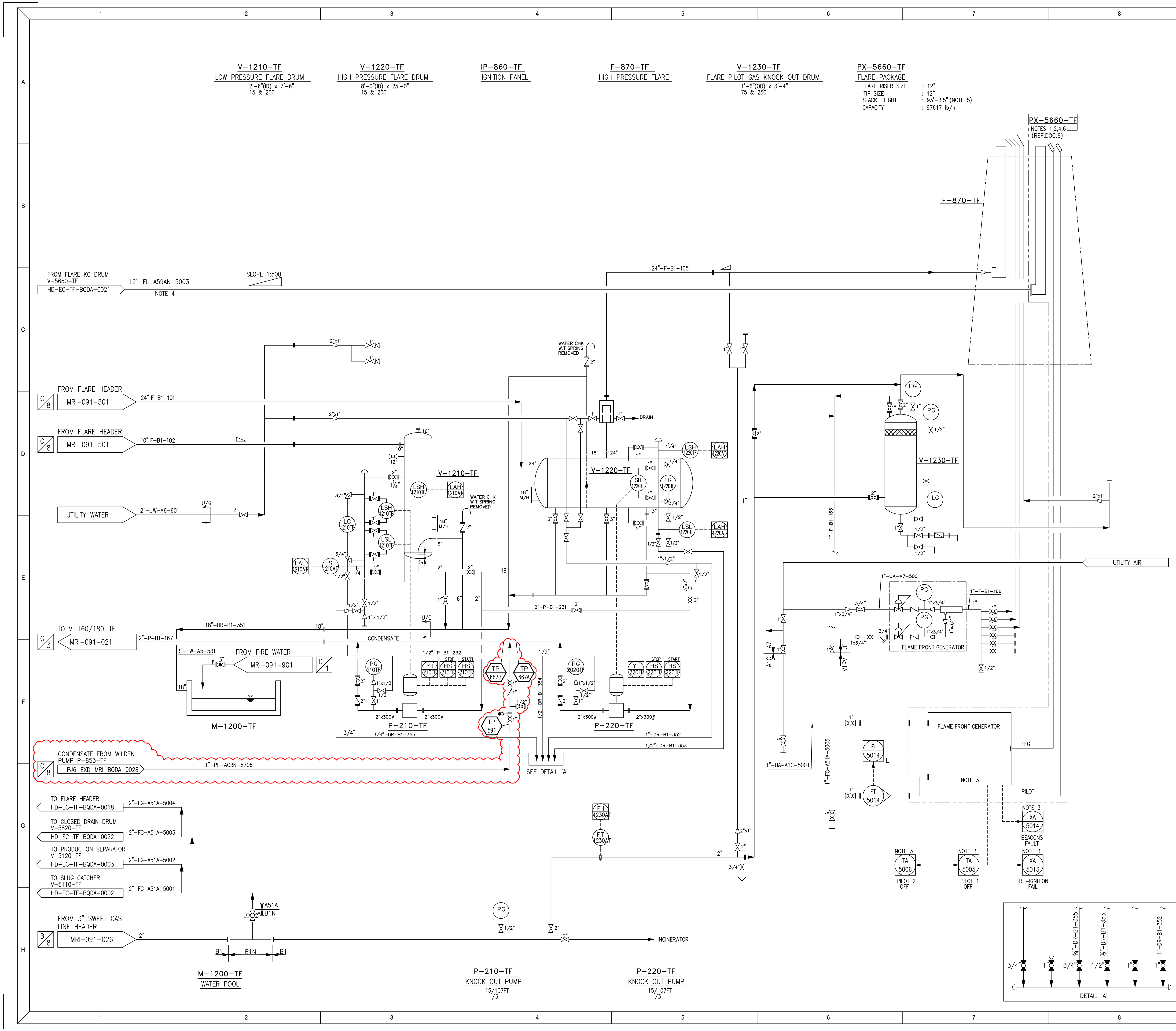
REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	PFD MUBARRAZ ISLAND (2/2)	PJ6-EXD-MRI-BQDA-0006
3	P&ID - MUBARRAZ TERMINAL SWEET GAS COMPRESSOR (K-101D-TF) (DEMOLITION)	PJ6-EXD-MRI-BQDA-0056

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
0	12/12/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
A	05/12/2025	ISSUED FOR REVIEW	TIM	MCS	PKR
REV.	DATE	DESCRIPTION	DR'N	CHD	AP'D
SCALE:	NTS	LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002		
ENGINEERING CONSULTANT: AREJLERS REJLERS TOWER 13TH FLOOR HAMDAN STREET, PO BOX 39317 ABU DHABI, UAE					
ENG CONSULTANT PROJECT NO: 5901055					
PMC : DORSCH HOLDING GmbH - ABU DHABI					
COMPANY: ABU DHABI OIL CO., LTD.					
PROJECT: DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)					
DRG. TITLE: PIPING AND INSTRUMENTATION DIAGRAM MUBARRAZ TERMINAL SWEET GAS COMPRESSOR (K-101D-TF)					
DWG. NO.	PJ6-EXD-MRI-BQDA-0055		REV NO.	SHEET NO.	
			1	001	

Filename : PJ6-EXD-MRI-BQDA-0055_1.DWG

SHEET SIZE: A1



NOTES

1. FLARE PILOT IS CONTINUOUS LIT TYPE, IGNITION SYSTEM IS FLAME FRONT GENERATOR.

2. TEMPERATURE ELEMENTS ARE THERMOCOUPLE TYPE.

3. THESE SIGNALS AND INDICATIONS WILL BE PROVIDED BY VENDOR ON PANEL AND WILL BE VISIBLE ON PCS.

4. FLARE SYSTEM (PSV'S, SIZE OF SUCTION AND DISCHARGE LINES, FLARE HEADERS, KO DRUM AND PUMPS) AND FLARE PACKAGE ARE DESIGNED ACCORDING TO FEED DESIGN CRITERIA (FOR REFERENCE SEE LETTERS HDS-EPC-L-T-164 FLARE HEADER, FLARE KNOCK-OUT DRUM AND FLARE KNOCK-OUT DRUM PUMP CAPACITY AND 8598-TRAD-ADOC-LET-PM-0004 EARLY PROCUREMENT OF FLARE PACKAGE).

5. STACK HEIGHT INDICATED IS BASED ON EXISTING FLARE DRAWING UP TO FLARE TIP.

6. FOR DETAILED INFORMATION SEE REFERENCE DOCUMENT 6.

7. EARLY TIE-IN(TP-667A/B) SHALL BE EXECUTED DURING YEAR 2026 SHUTDOWN.

GENERAL NOTES:

1. FOR DEMOLITION SCOPE REFER TO P&ID NO. HD-EC-TF-BQDA-0023.

2. ALL INSTRUMENT TAG NUMBERS WILL BE SUFFIXED WITH "T" FOR MUBARRAZ ISLAND.

3. PIPING TIE-IN DETAILS ARRANGEMENTS HAVE BEEN MODIFIED ACCORDING TO LETTER HDS-EPC-L-T-016 PIPING TIE-IN DETAILS AND 8598-TRAD-ADOC-MOM-PM-0065 3RD CONSTRUCTION SITE VISIT WRAP UP MEETING.

4. ONLY ELEMENTS INSIDE CLOUD ARE SCOPE OF HAIL SURFACE FACILITIES EPS PROJECT. REST OF INFO IS SHOWN ONLY FOR INFORMATION (ADOC SCOPE).

5. FOR TIE-IN CONNECTION TYPE, PIPING CLASS AND PIPING DETAIL REFER TO TIE-IN LIST DOC NO. HD-EC-CM-BQOL-0002.

6. INSULATION KIT SHALL BE PROVIDED FOR DISSIMILAR METAL JOINTS.

7. FOR AFTER TIE-IN SCOPE REFER TO P&ID HD-EC-TF-BQDA-0044.

LEGEND

SCOPE CLOUD

TIE-IN POINT

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGENDS AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002
2	P&ID MUBARRAZ TERMINAL FLARE SYSTEM (2/2) (DEMOLITION)	PJ6-EXD-MRI-BQDA-0059

THIS DRAWING HAS BEEN DEVELOPED FROM PROJECT NO.8598 DRAWING NO.HD-EC-TF-BQDA-0024 Rev 3. CLOUD PORTION DENOTES SCOPE UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"

REV.	DATE	DESCRIPTION	DR'N	CHD	AP'D
1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS	PKR
0	06/01/2026	APPROVED FOR CONSTRUCTION	DVS	MCS	PKR

SCALE:	LOCATION:	COMPANY PROJECT NO.
NTS	MUBARRAZ ISLAND	PJ-2025-CR-002

ENGINEERING CONSULTANT:

REJLERS

REJLERS TOWER 13TH FLOOR
HAMDAN STREET,PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO:

5901055

PMC:

DORSCH HOLDING GmbH - ABU DHABI

COMPANY:

ABU DHABI OIL CO.,LTD.

PROJECT:

DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)

DRG. TITLE:

PIPING AND INSTRUMENTATION DIAGRAM
MUBARRAZ TERMINAL FLARE SYSTEM (2/2)

DWG. NO.	PJ6-EXD-MRI-BQDA-0057	REV NO.	1	SHEET NO.	001
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Filename : PJ6-EXD-MRI-BQDA-0057_1.DWG

SHEET SIZE: A1

P-8005A/B-TF
CHEMICAL TREATING PUMP

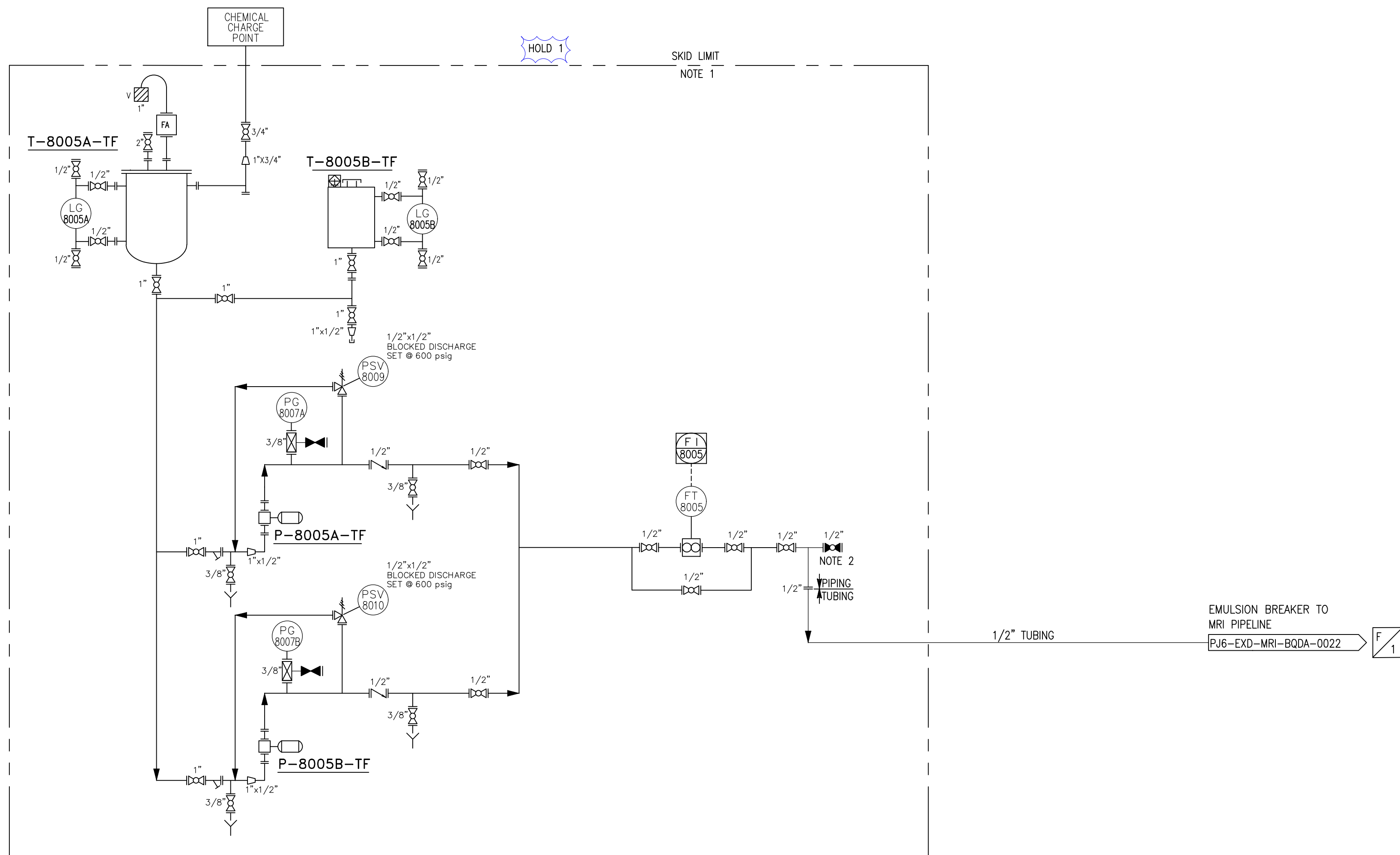
RATED FLOW	: 60 l/hr
RATED DIFFERENTIAL PRESSURE/HEAD	: 174.1 psig/426.5 ft
DISCHARGE PRESSURE/DESIGN PRESSURE	: 174 psig /600 psig
HYDRAULIC POWER	: 2.0 kW
TYPE	: DIAPHRAGM

T-8005A-TF
CHEMICAL STORAGE TANK

NOMINAL CAPACITY : 9.25 m3
DIAMETER : 1.95 m
HEIGHT (T/T) : 3.1 m
OPERATING PRESS. : ATM

T-8005B-TF
TEST RUN TANK

NOMINAL CAPACITY	: 1.2 m ³
LENGTH	: 1 m
WIDTH	: 1 m
HEIGHT	: 1.2 m
OPERATING PRESS.	: ATM



NOTES

1. ALL INSTRUMENT TAG NUMBERS ARE SUFFIXED WITH "TF" FOR MUBARRAZ ISLAND.
2. FOR FUTURE CONNECTION.
3. CHEMICAL INJECTION SYSTEM SHOWN IS MINIMUM REQUIREMENT. REFER VENDOR P&ID FOR FURTHER DETAILS.

LEGEND

- | | |
|---|----------------|
|  | SCOPE CLOUD |
|  | REVISION CLOUD |
|  | HOLD CLOUD |

HOLD

1. MRD CHEMICAL INJECTION PACKAGE VENDOR P&ID DETAILS.

REFERENCE DOCUMENTS / DRAWINGS

S.No.	DOCUMENT / DRAWING TITLE	DOC / DWG No.
1	P&ID LEGEND AND SYMBOLS	PJ6-EXD-GEN-BQDA-0002

1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
0	12/12/2025	APPROVED FOR CONSTRUCTION	TIM	MCS	PKR
REV.	DATE	DESCRIPTION	DR'N	CH'D	AP'D
SCALE: NTS	LOCATION: MUBARRAZ ISLAND		COMPANY PROJECT NO. PJ-2025-CR-002		

ENGINEERING CONSULTANT:

REJLERS

REJLERS TOWER 13TH FLOOR
HAMDAN STREET, PO BOX 39317
ABU DHABI, UAE

ENG CONSULTANT PROJECT NO: 5901055

PMC : DORSCH HOLDING GmbH - ABU DHAB

COMPANY:

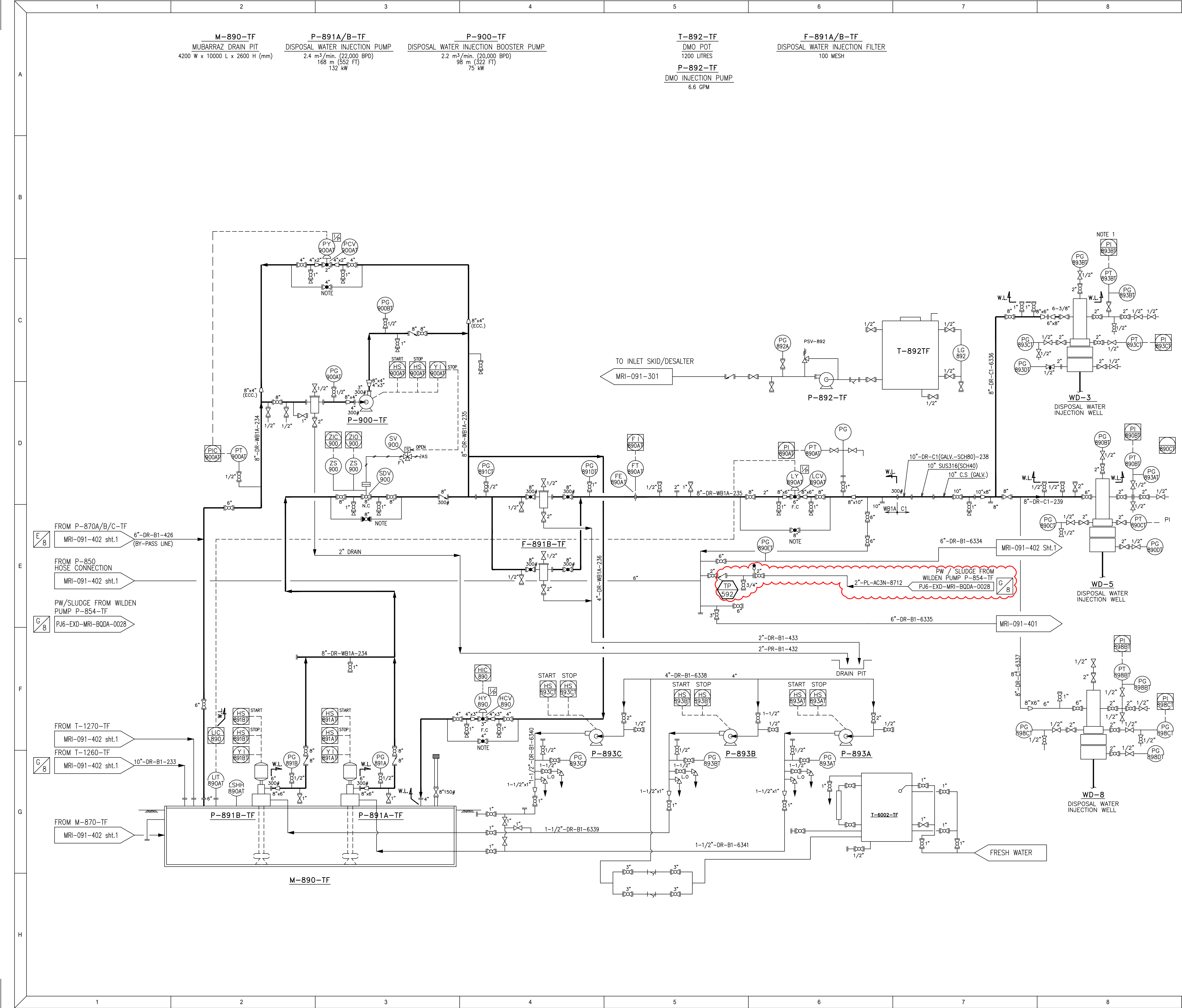


ABU DHABI OIL CO., LTD.

PROJECT:	DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)
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DRG. TITLE:	PIPING AND INSTRUMENTATION DIAGRAM - MUBARRAZ TERMINAL - MRD CHEMICAL INJECTION SYSTEM
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DWG. NO.	PJ6-EXD-MRI-BQDA-0058	REV NO.	SHEET NO
		1	001



NOTES				
1. IT IS NECESSARY THAT THE DISCUSSION OF THE SPECIFICATION OF THE INSTRUMENT EQUIPMENT WITH THE MANUFACTURE BASED ON THIS STUDY DATA AT THE PURCHASE STAGE.				
LEGEND				
		 TIE-IN POINT		
REFERENCE DOCUMENTS / DRAWINGS				
S.No.	DOCUMENT / DRAWING TITLE		DOC / DWG No.	
1	P&ID LEGENDS AND SYMBOLS		PJ6-EXD-GEN-BQDA-0002	
2	P&ID MUBARRAZ TERMINAL DRAIN SYSTEM SKIMMER PIT & INJECTION SYSTEM (2/2) (DEMOLITION)		PJ6-EXD-MRI-BQDA-0061	
THIS DRAWING HAS BEEN DEVELOPED FROM DRAWING NO. MRI-091-402 Rev 15. CLOUD PORTION DENOTES SCOPE UNDER DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT "PJ-2025-CR-002"				
				
1	30/01/2026	RE-APPROVED FOR CONSTRUCTION	DVS	MCS
0	06/01/2026	APPROVED FOR CONSTRUCTION	DVS	MCS
REV.	DATE	DESCRIPTION	DR'N	CH'D
SCALE: NTS		LOCATION: MUBARRAZ ISLAND	COMPANY PROJECT NO. PJ-2025-CR-002	
ENGINEERING CONSULTANT: REJLERS TOWER 13TH FLOOR HAMDAN STREET, PO BOX 39317 ABU DHABI, UAE				
ENG CONSULTANT PROJECT NO: 5901055				
PMC : DORSCH HOLDING GmbH - ABU DHABI				
COMPANY:  ABU DHABI OIL CO., LTD.				
PROJECT: DETAILED ENGINEERING FOR MUBARRAZ REDEVELOPMENT PROJECT (EPC PHASE)				
DRG. TITLE: PIPING AND INSTRUMENTATION DIAGRAM MUBARRAZ TERMINAL DRAIN SYSTEM SKIMMER PIT & INJECTION SYSTEM (2/2)				
DWG. NO.	PJ6-EXD-MRI-BQDA-0060		REV NO.	SHEET NO.
			1	001

File name : PJ6-EXD-MRI-BQDA-0060_1.DWG

